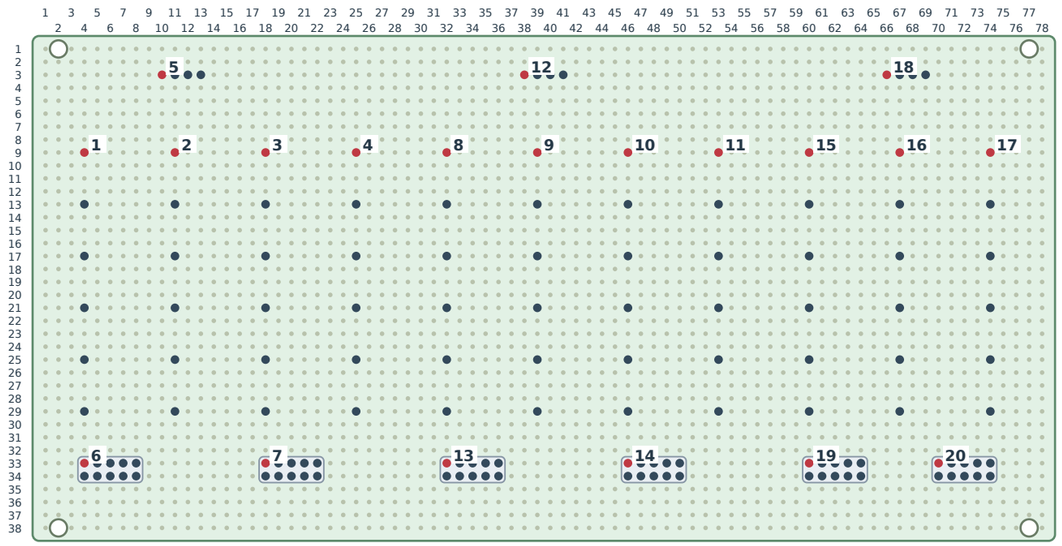
contact-a — placement

TOP coordinates retained on MIRRORED UNDERSIDE. Numbered reference tables follow.

contact-a — TOP

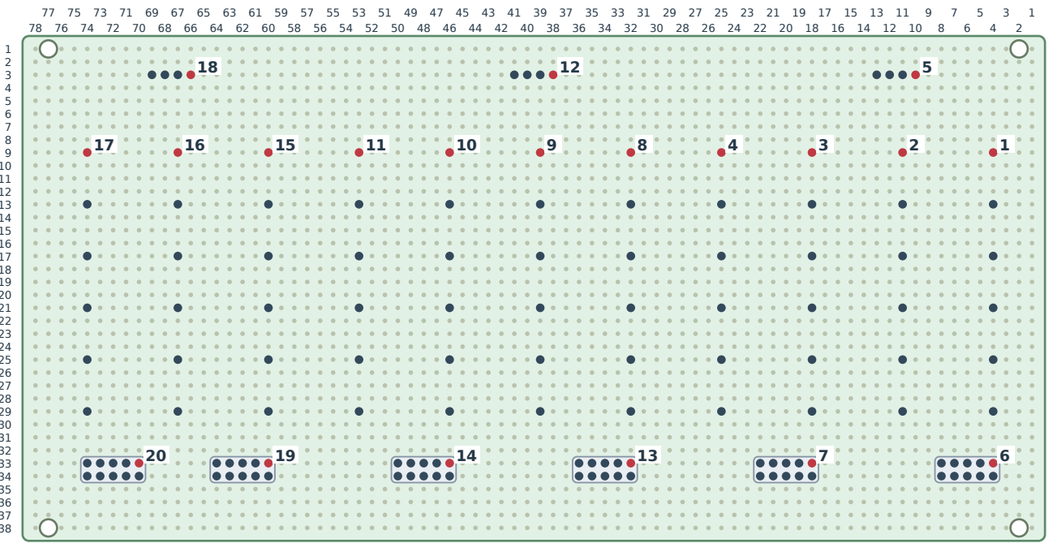


200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

contact-a — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

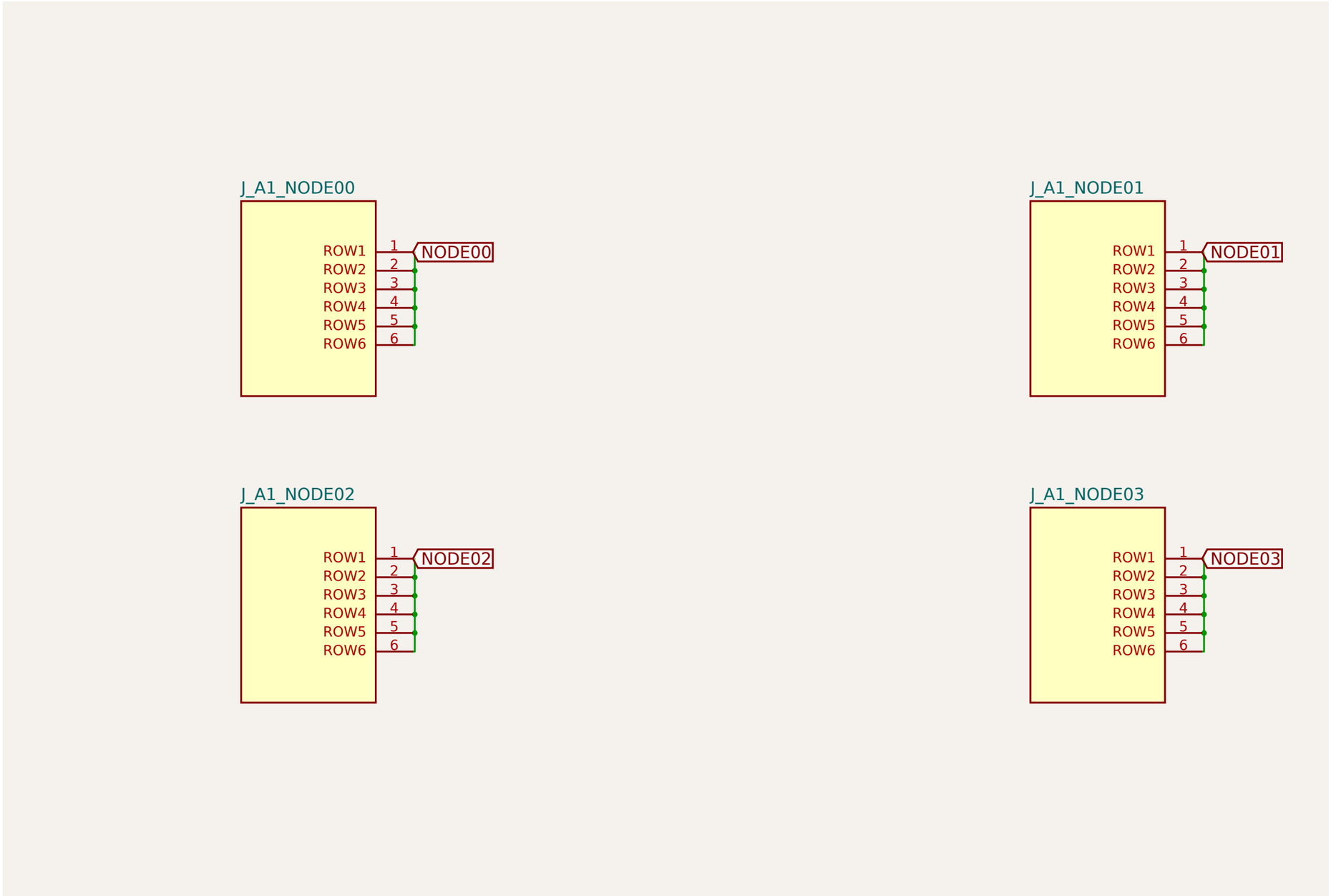
PROTOTYPE: physical fit and hardware tests pending.

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

contact-a — components and endpoints

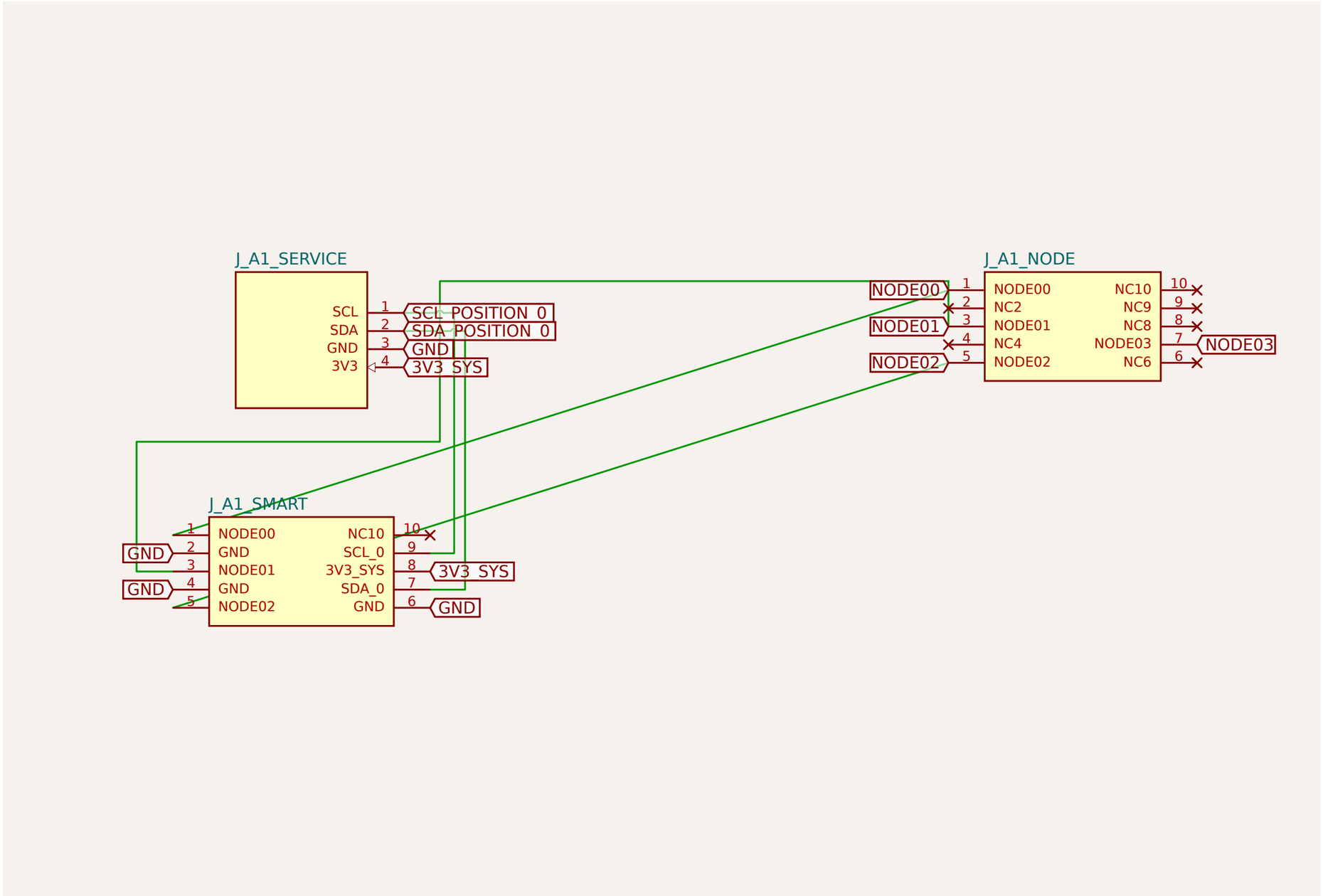
Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
1	J_A1_NODE00: NODE00 — six contacts	CP00	1=4,9; 2=4,13; 3=4,17; 4=4,21; 5=4,25; 6=4,29	
2	J_A1_NODE01: NODE01 — six contacts	CP00	1=11,9; 2=11,13; 3=11,17; 4=11,21; 5=11,25; 6=11,29	
3	J_A1_NODE02: NODE02 — six contacts	CP00	1=18,9; 2=18,13; 3=18,17; 4=18,21; 5=18,25; 6=18,29	
4	J_A1_NODE03: NODE03 — six contacts	CP00	1=25,9; 2=25,13; 3=25,17; 4=25,21; 5=25,25; 6=25,29	
5	J_A1_SERVICE: OLED service pogo contacts	CP06	1=10,3; 2=11,3; 3=12,3; 4=13,3	
6	J_A1_NODE: NODE cable → matrix	CP00	1=4,33; 2=4,34; 3=5,33; 4=5,34; 5=6,33; 6=6,34; 7=7,33; 8=7,34; 9=8,33; 10=8,34	
7	J_A1_SMART: SMART_RAW v2 → channel 0	CP06	1=18,33; 2=18,34; 3=19,33; 4=19,34; 5=20,33; 6=20,34; 7=21,33; 8=21,34; 9=22,33; 10=22,34	
8	J_A2_NODE04: NODE04 — six contacts	CP27	1=32,9; 2=32,13; 3=32,17; 4=32,21; 5=32,25; 6=32,29	
9	J_A2_NODE05: NODE05 — six contacts	CP27	1=39,9; 2=39,13; 3=39,17; 4=39,21; 5=39,25; 6=39,29	
10	J_A2_NODE06: NODE06 — six contacts	CP27	1=46,9; 2=46,13; 3=46,17; 4=46,21; 5=46,25; 6=46,29	
11	J_A2_NODE07: NODE07 — six contacts	CP27	1=53,9; 2=53,13; 3=53,17; 4=53,21; 5=53,25; 6=53,29	
12	J_A2_SERVICE: OLED service pogo contacts	CP27	1=38,3; 2=39,3; 3=40,3; 4=41,3	
13	J_A2_NODE: NODE cable → matrix	CP27	1=32,33; 2=32,34; 3=33,33; 4=33,34; 5=34,33; 6=34,34; 7=35,33; 8=35,34; 9=36,33; 10=36,34	
14	J_A2_SMART: SMART_RAW v2 → channel 1	CP27	1=46,33; 2=46,34; 3=47,33; 4=47,34; 5=48,33; 6=48,34; 7=49,33; 8=49,34; 9=50,33; 10=50,34	
15	J_A3_NODE08: NODE08 — six contacts	CP28	1=60,9; 2=60,13; 3=60,17; 4=60,21; 5=60,25; 6=60,29	
16	J_A3_NODE09: NODE09 — six contacts	CP28	1=67,9; 2=67,13; 3=67,17; 4=67,21; 5=67,25; 6=67,29	
17	J_A3_NODE10: NODE10 — six contacts	CP28	1=74,9; 2=74,13; 3=74,17; 4=74,21; 5=74,25; 6=74,29	
18	J_A3_SERVICE: OLED service pogo contacts	CP28	1=66,3; 2=67,3; 3=68,3; 4=69,3	
19	J_A3_NODE: NODE cable → matrix	CP28	1=60,33; 2=60,34; 3=61,33; 4=61,34; 5=62,33; 6=62,34; 7=63,33; 8=63,34; 9=64,33; 10=64,34	
20	J_A3_SMART: SMART_RAW v2 → channel 2	CP28	1=70,33; 2=70,34; 3=71,33; 4=71,34; 5=72,33; 6=72,34; 7=73,33; 8=73,34; 9=74,33; 10=74,34	

contact-a — a1: Nodes



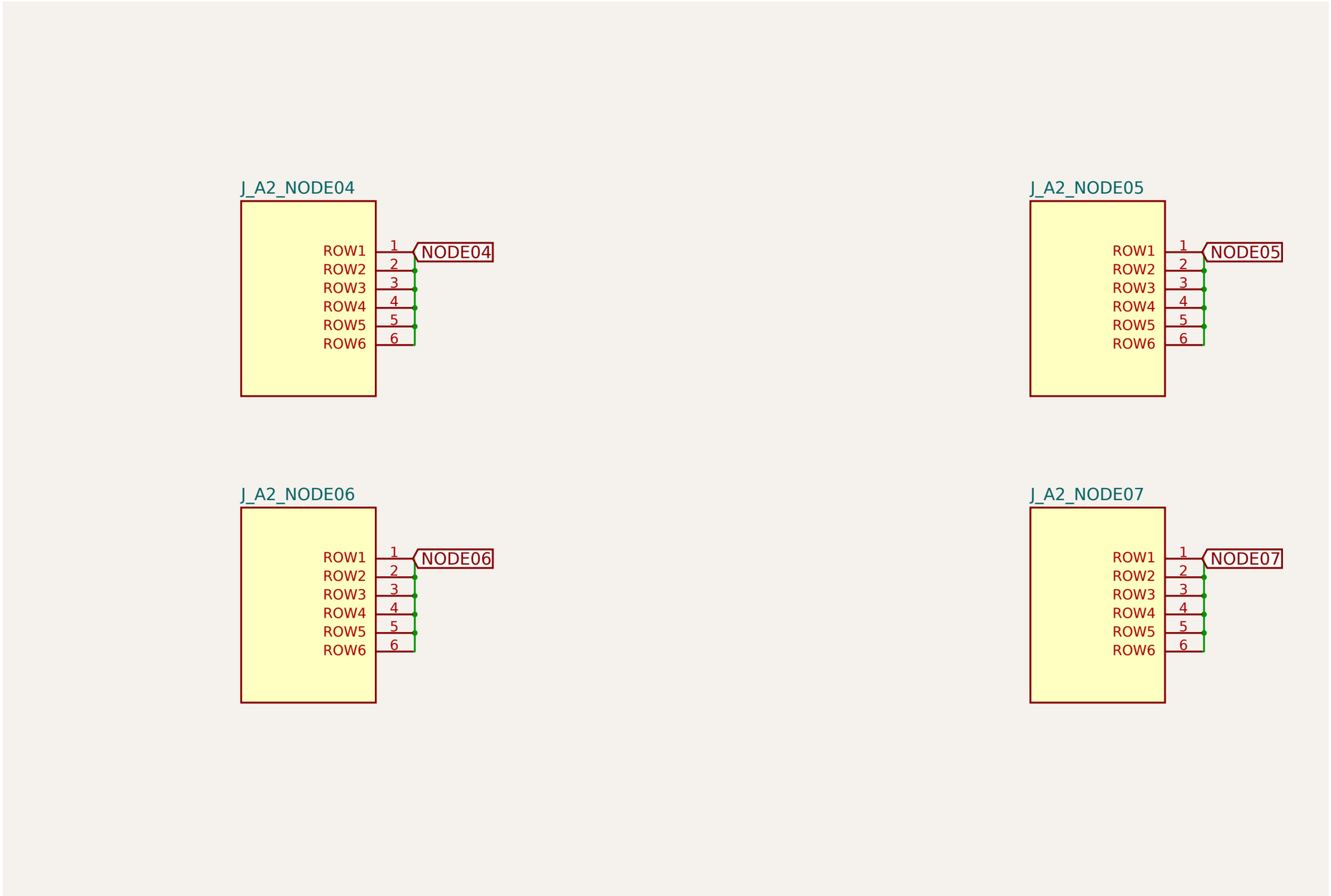
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-a — a1: Service



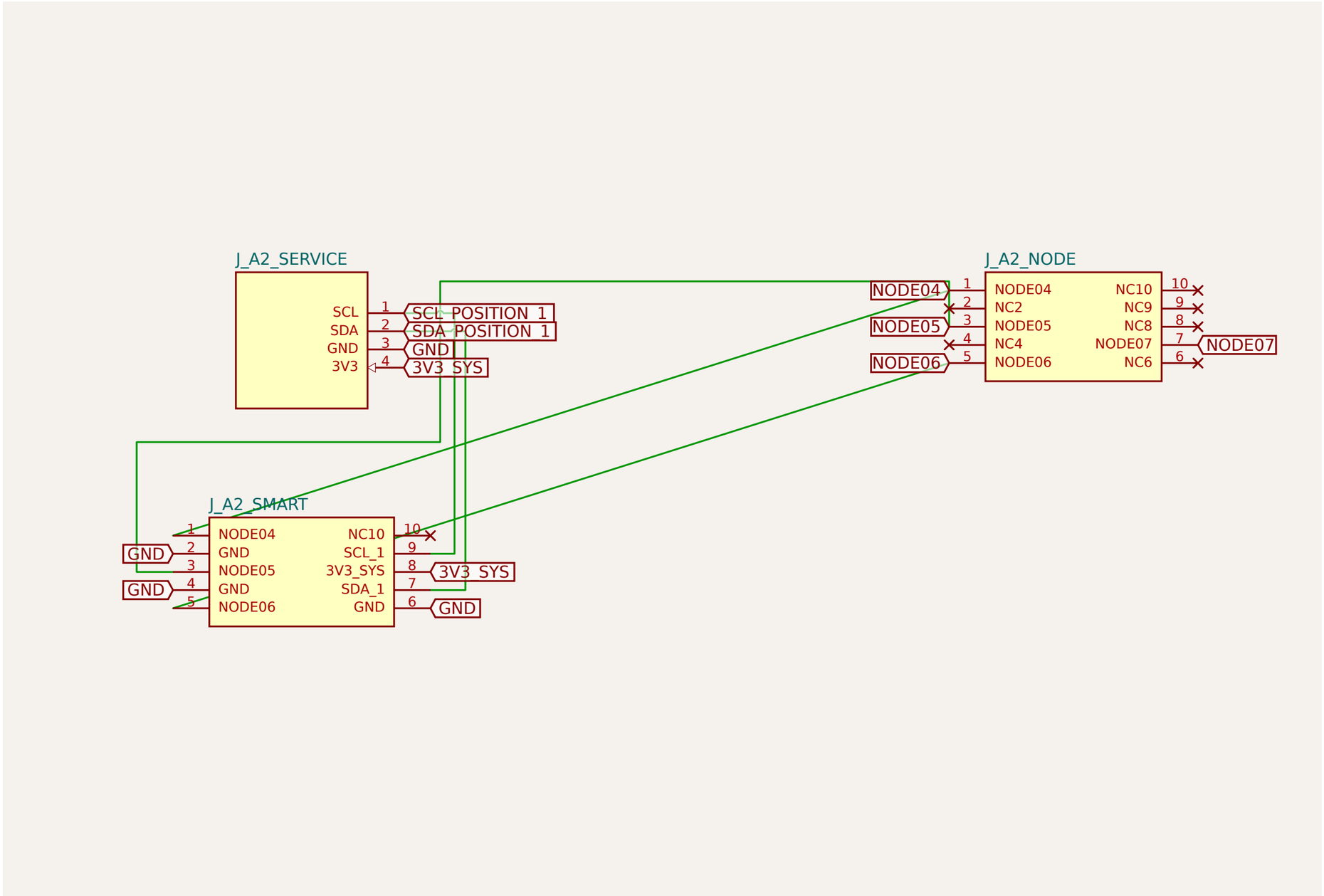
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-a — a2: Nodes



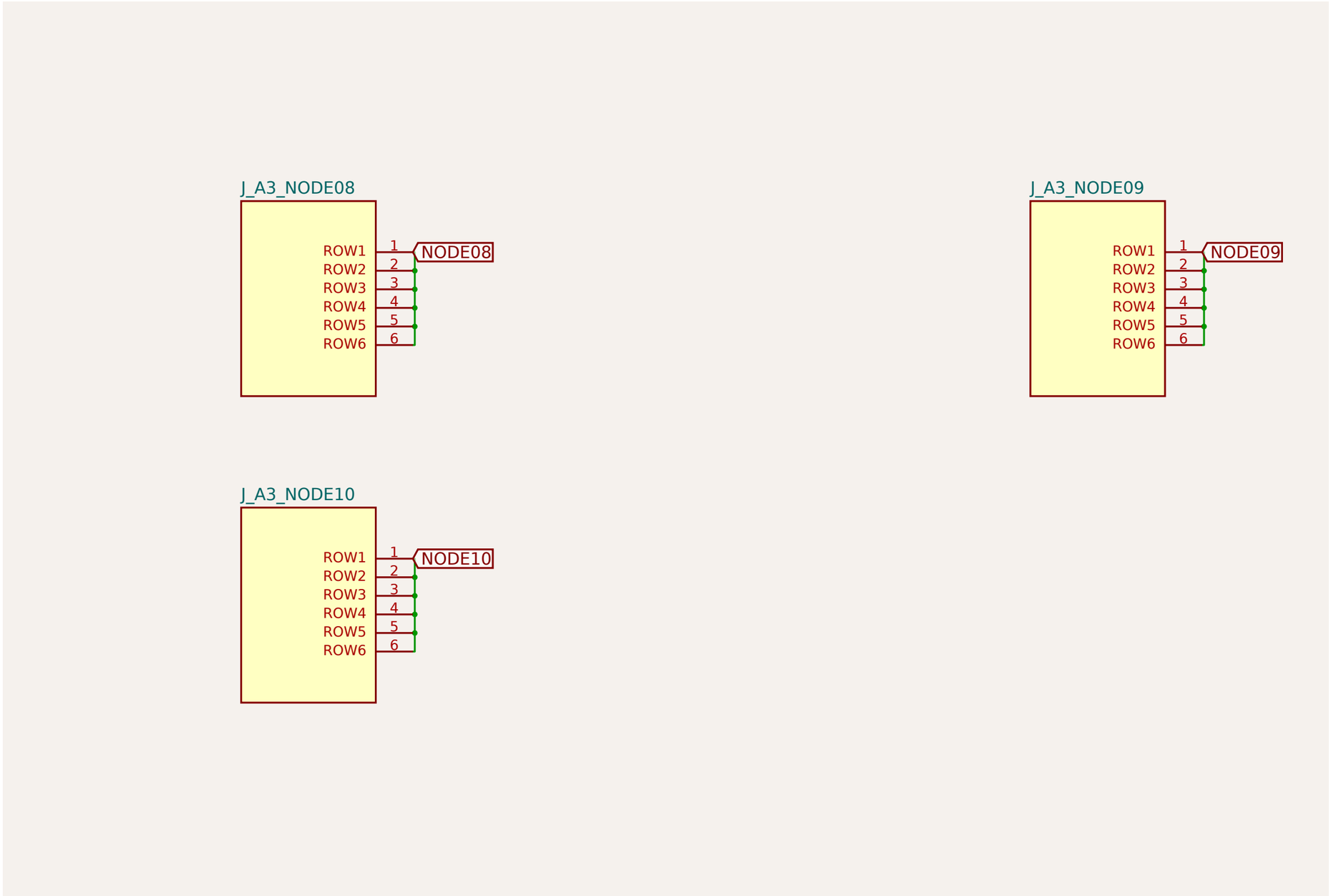
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-a — a2: Service



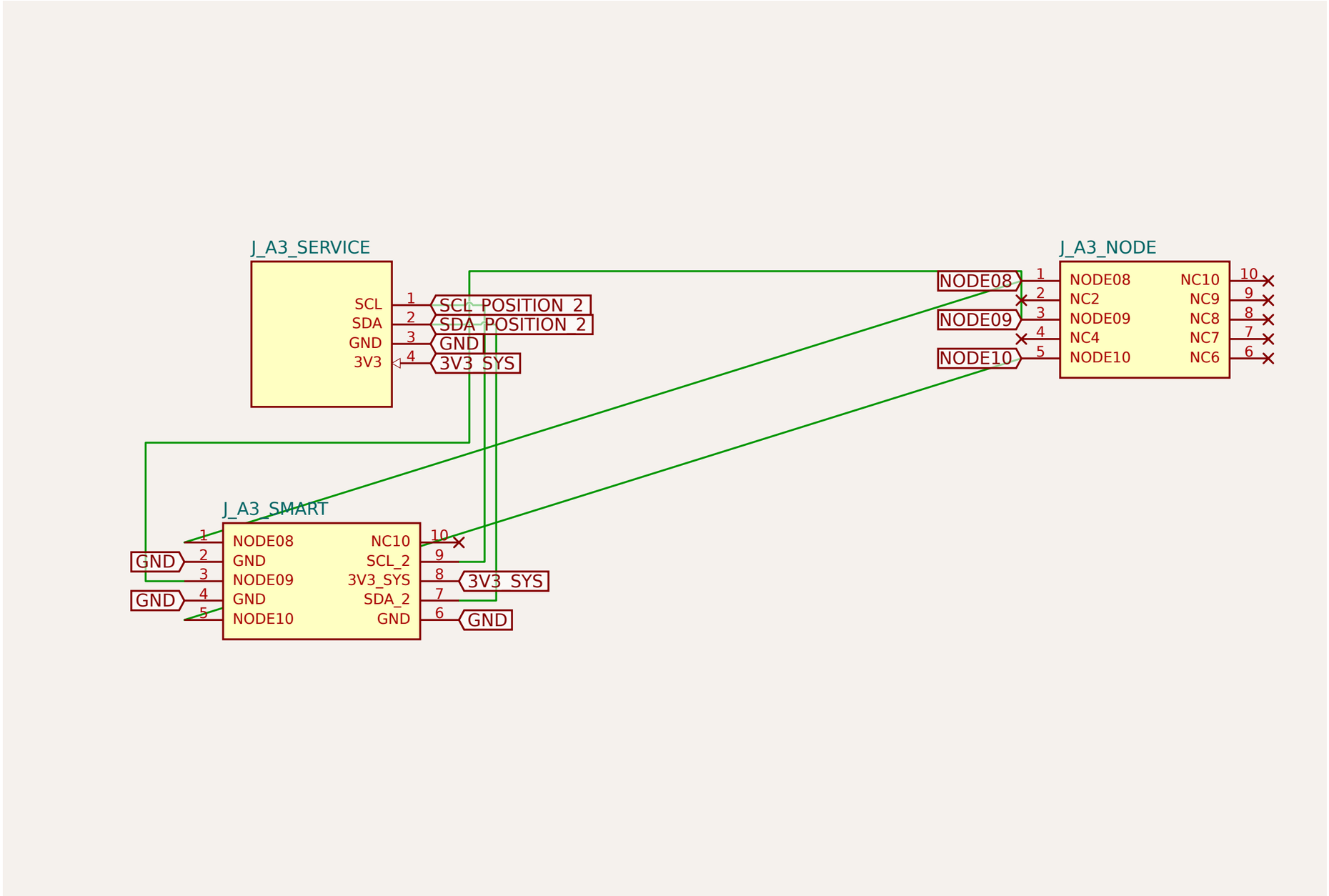
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-a — a3: Nodes



Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-a — a3: Service

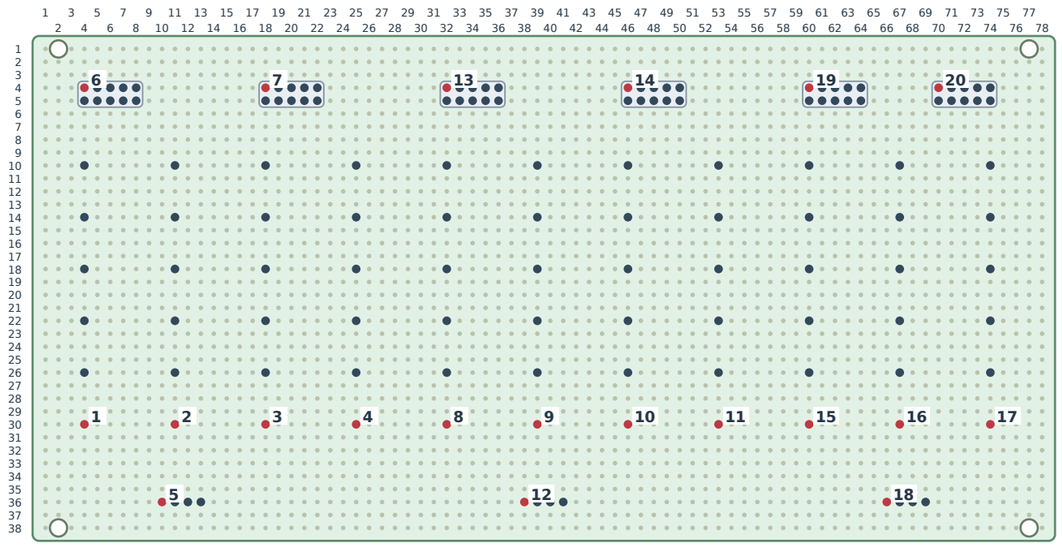


Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — placement

TOP coordinates retained on MIRRORED UNDERSIDE. Numbered reference tables follow.

contact-b — TOP

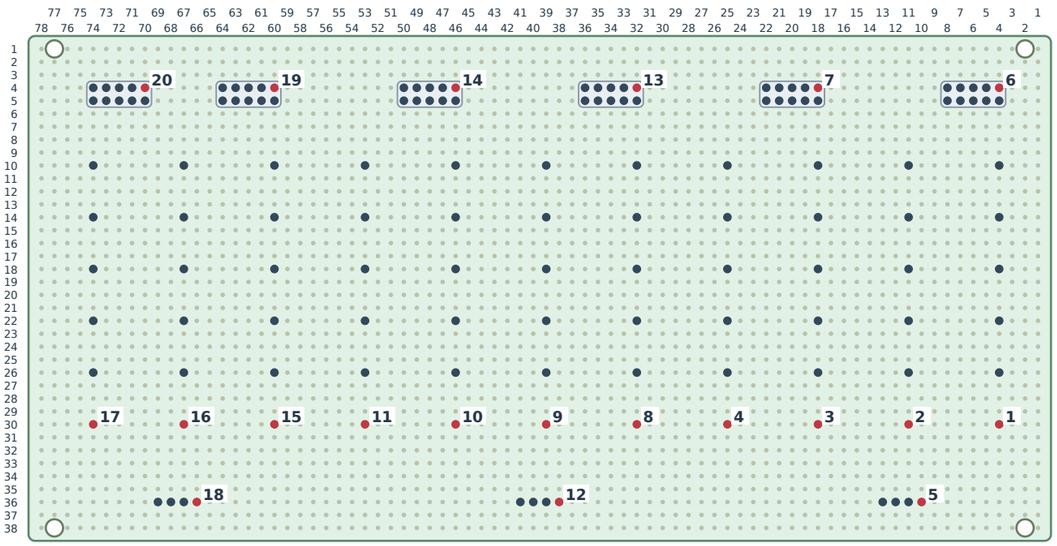


200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

contact-b — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

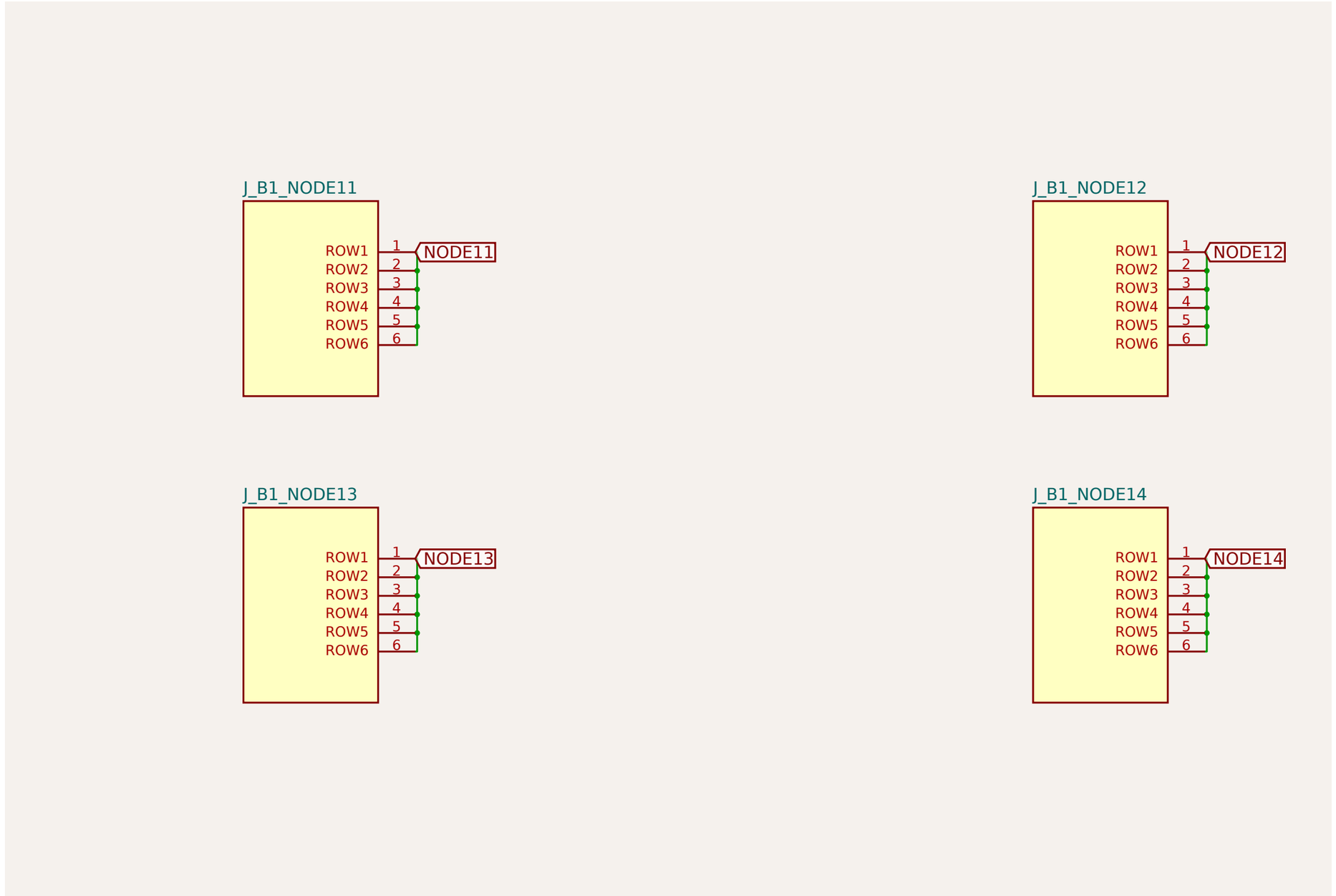
PROTOTYPE: physical fit and hardware tests pending.

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

contact-b — components and endpoints

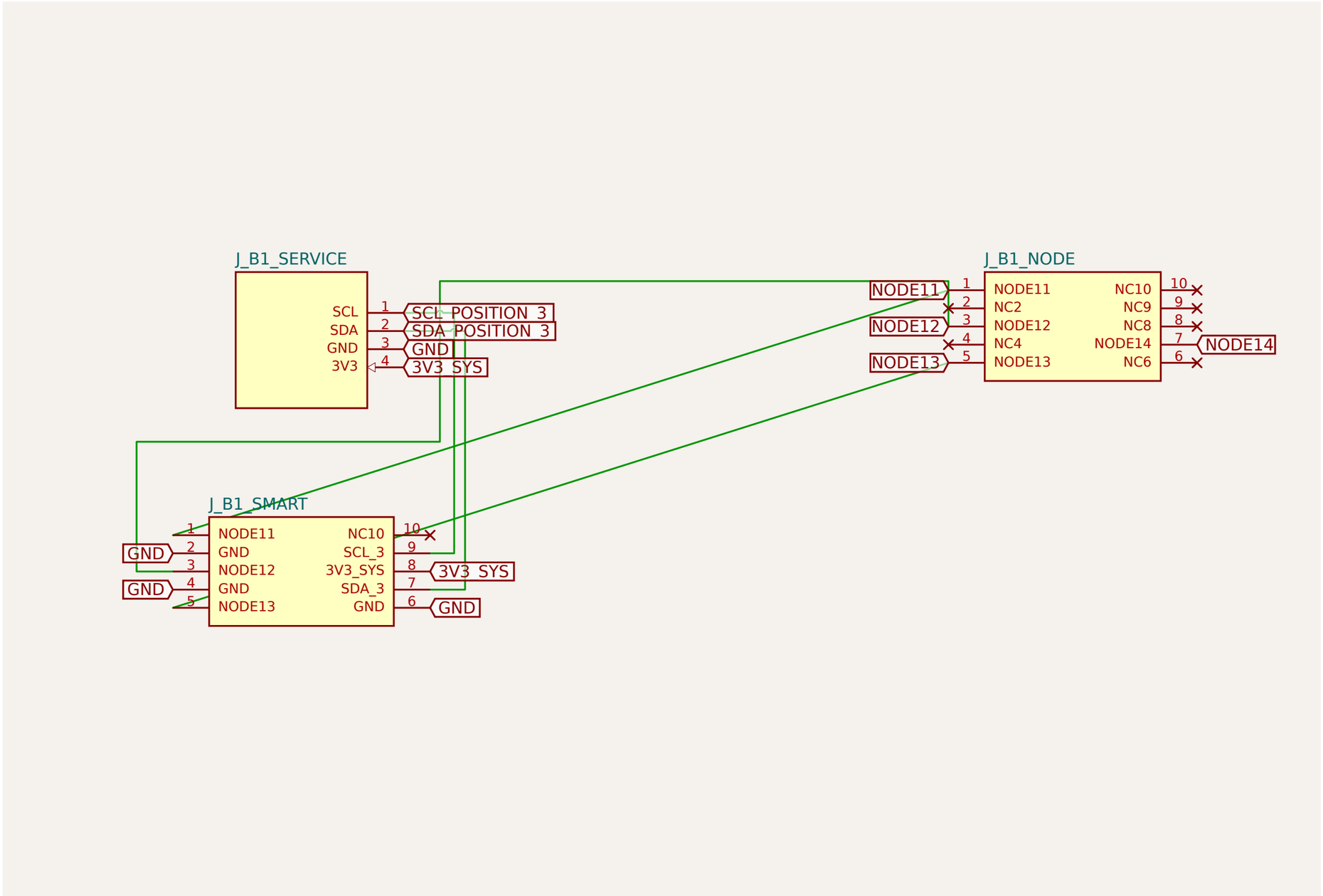
Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
1	J_B1_NODE11: NODE11 — six contacts	CP29	1=4,30; 2=4,26; 3=4,22; 4=4,18; 5=4,14; 6=4,10	
2	J_B1_NODE12: NODE12 — six contacts	CP29	1=11,30; 2=11,26; 3=11,22; 4=11,18; 5=11,14; 6=11,10	
3	J_B1_NODE13: NODE13 — six contacts	CP29	1=18,30; 2=18,26; 3=18,22; 4=18,18; 5=18,14; 6=18,10	
4	J_B1_NODE14: NODE14 — six contacts	CP29	1=25,30; 2=25,26; 3=25,22; 4=25,18; 5=25,14; 6=25,10	
5	J_B1_SERVICE: OLED service pogo contacts	CP29	1=10,36; 2=11,36; 3=12,36; 4=13,36	
6	J_B1_NODE: NODE cable → matrix	CP29	1=4,4; 2=4,5; 3=5,4; 4=5,5; 5=6,4; 6=6,5; 7=7,4; 8=7,5; 9=8,4; 10=8,5	
7	J_B1_SMART: SMART_RAW v2 → channel 3	CP29	1=18,4; 2=18,5; 3=19,4; 4=19,5; 5=20,4; 6=20,5; 7=21,4; 8=21,5; 9=22,4; 10=22,5	
8	J_B2_NODE15: NODE15 — six contacts	CP30	1=32,30; 2=32,26; 3=32,22; 4=32,18; 5=32,14; 6=32,10	
9	J_B2_NODE16: NODE16 — six contacts	CP30	1=39,30; 2=39,26; 3=39,22; 4=39,18; 5=39,14; 6=39,10	
10	J_B2_NODE17: NODE17 — six contacts	CP30	1=46,30; 2=46,26; 3=46,22; 4=46,18; 5=46,14; 6=46,10	
11	J_B2_NODE18: NODE18 — six contacts	CP30	1=53,30; 2=53,26; 3=53,22; 4=53,18; 5=53,14; 6=53,10	
12	J_B2_SERVICE: OLED service pogo contacts	CP30	1=38,36; 2=39,36; 3=40,36; 4=41,36	
13	J_B2_NODE: NODE cable → matrix	CP30	1=32,4; 2=32,5; 3=33,4; 4=33,5; 5=34,4; 6=34,5; 7=35,4; 8=35,5; 9=36,4; 10=36,5	
14	J_B2_SMART: SMART_RAW v2 → channel 4	CP30	1=46,4; 2=46,5; 3=47,4; 4=47,5; 5=48,4; 6=48,5; 7=49,4; 8=49,5; 9=50,4; 10=50,5	
15	J_B3_NODE19: NODE19 — six contacts	CP31	1=60,30; 2=60,26; 3=60,22; 4=60,18; 5=60,14; 6=60,10	
16	J_B3_NODE20: NODE20 — six contacts	CP31	1=67,30; 2=67,26; 3=67,22; 4=67,18; 5=67,14; 6=67,10	
17	J_B3_NODE21: NODE21 — six contacts	CP31	1=74,30; 2=74,26; 3=74,22; 4=74,18; 5=74,14; 6=74,10	
18	J_B3_SERVICE: OLED service pogo contacts	CP31	1=66,36; 2=67,36; 3=68,36; 4=69,36	
19	J_B3_NODE: NODE cable → matrix	CP31	1=60,4; 2=60,5; 3=61,4; 4=61,5; 5=62,4; 6=62,5; 7=63,4; 8=63,5; 9=64,4; 10=64,5	
20	J_B3_SMART: SMART_RAW v2 → channel 5	CP31	1=70,4; 2=70,5; 3=71,4; 4=71,5; 5=72,4; 6=72,5; 7=73,4; 8=73,5; 9=74,4; 10=74,5	

contact-b — b1: Nodes



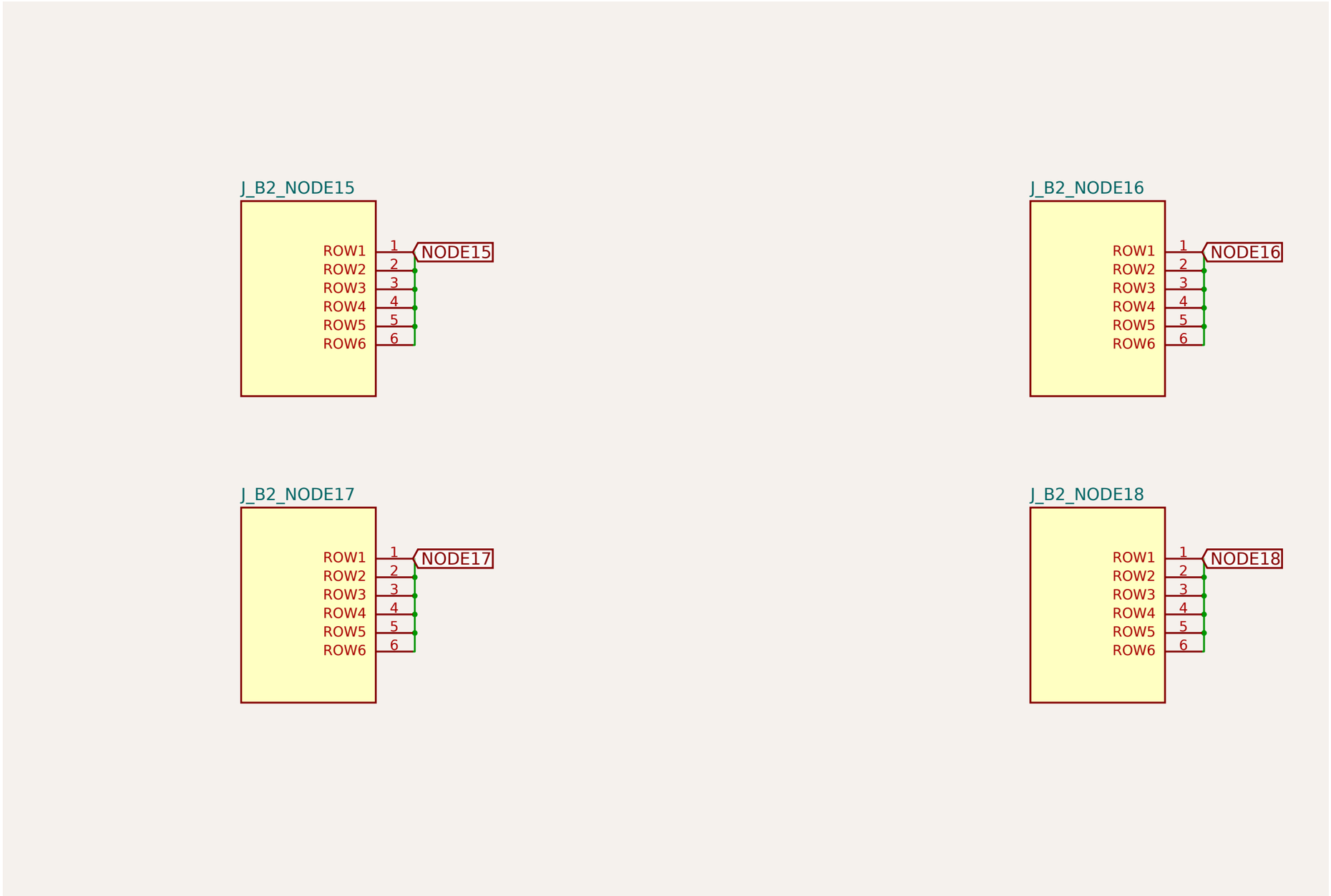
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — b1: Service



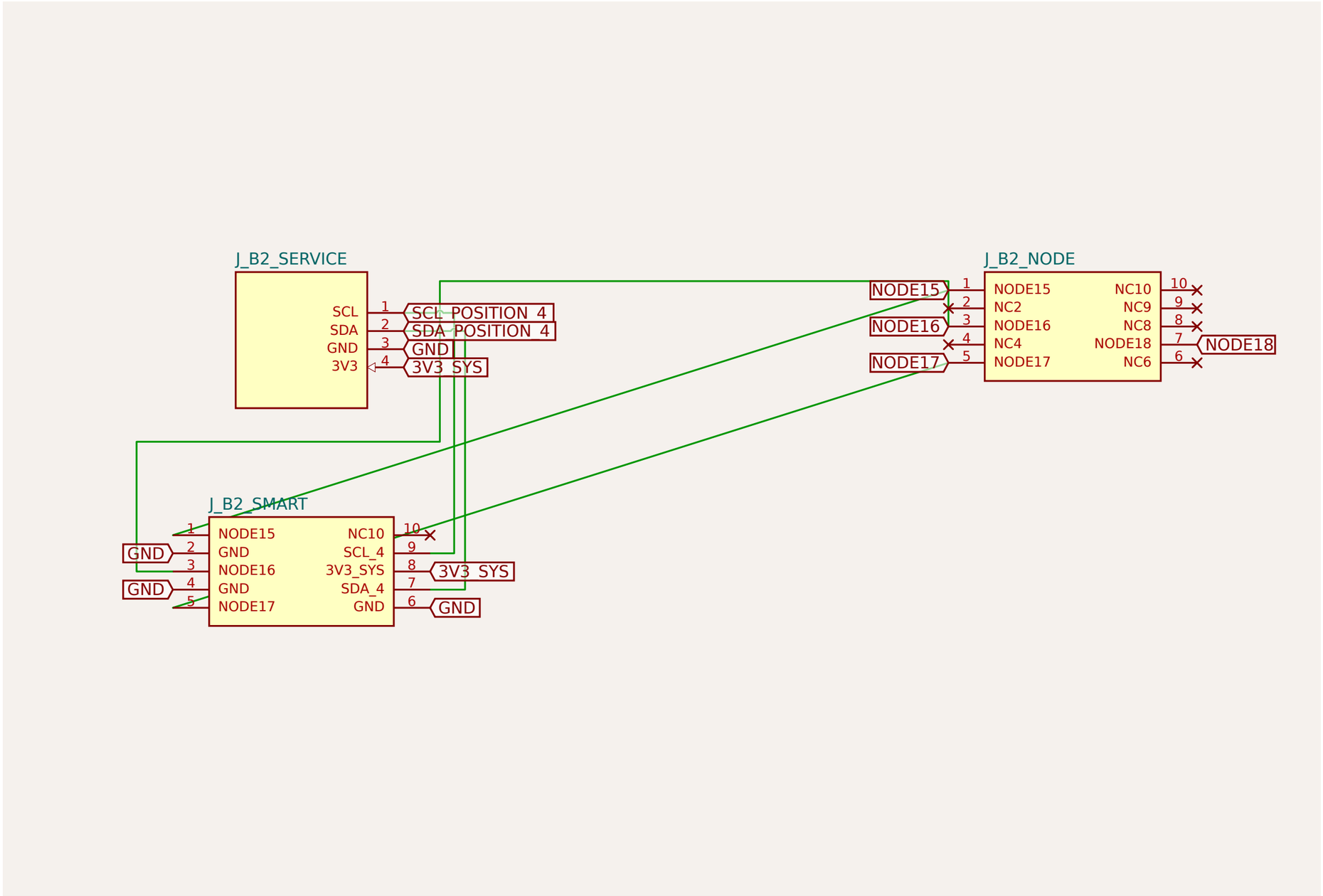
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — b2: Nodes



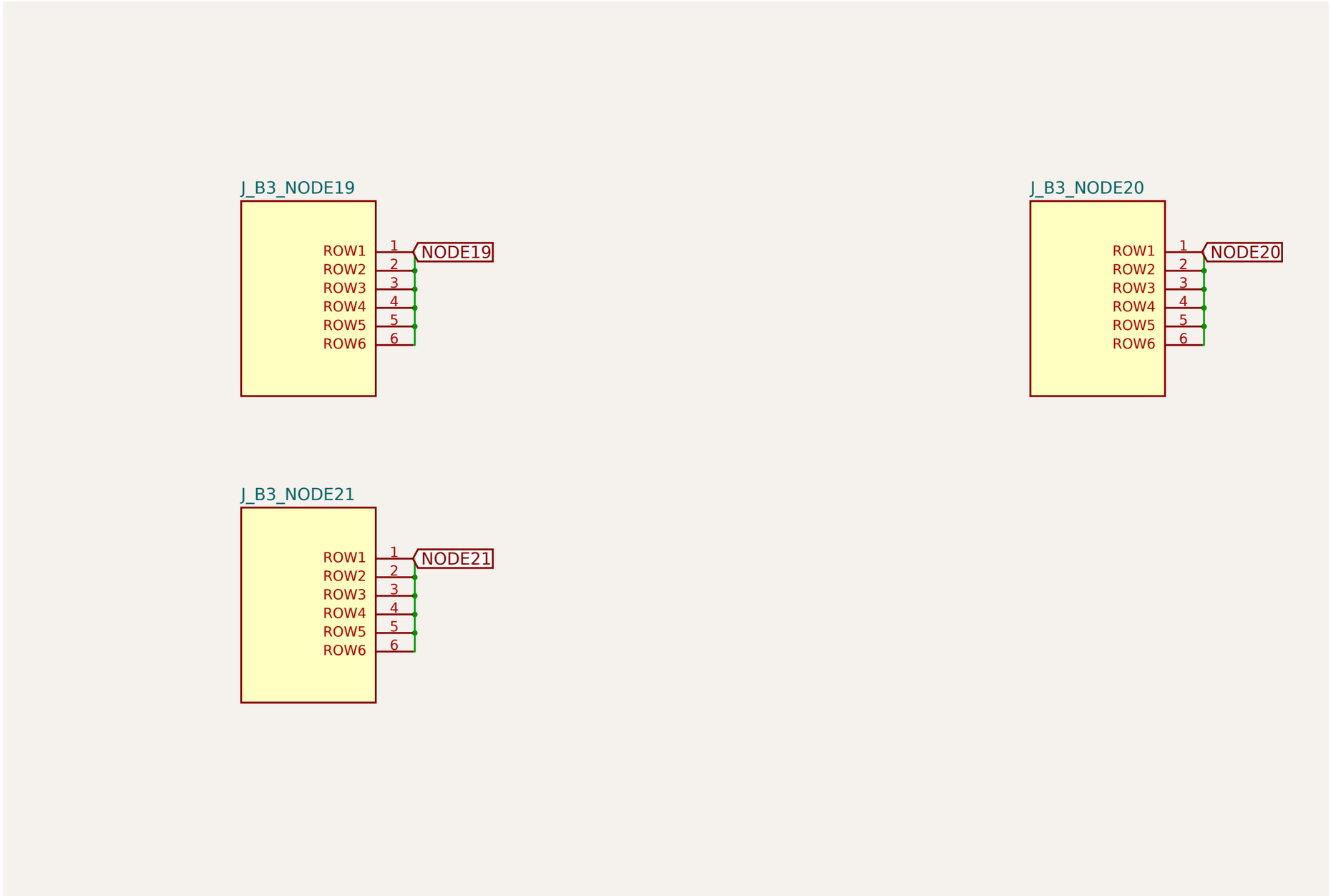
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — b2: Service



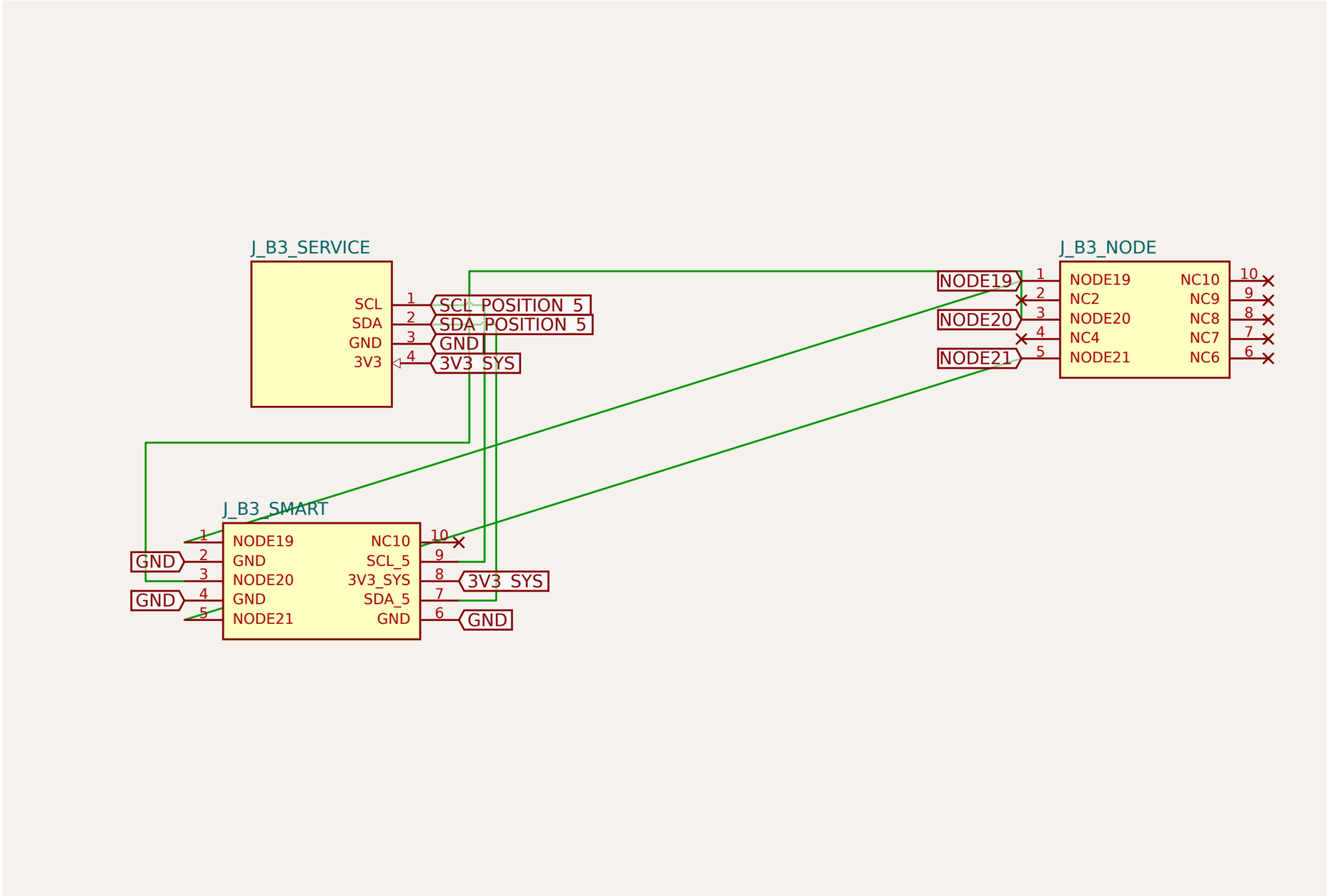
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — b3: Nodes



Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

contact-b — b3: Service

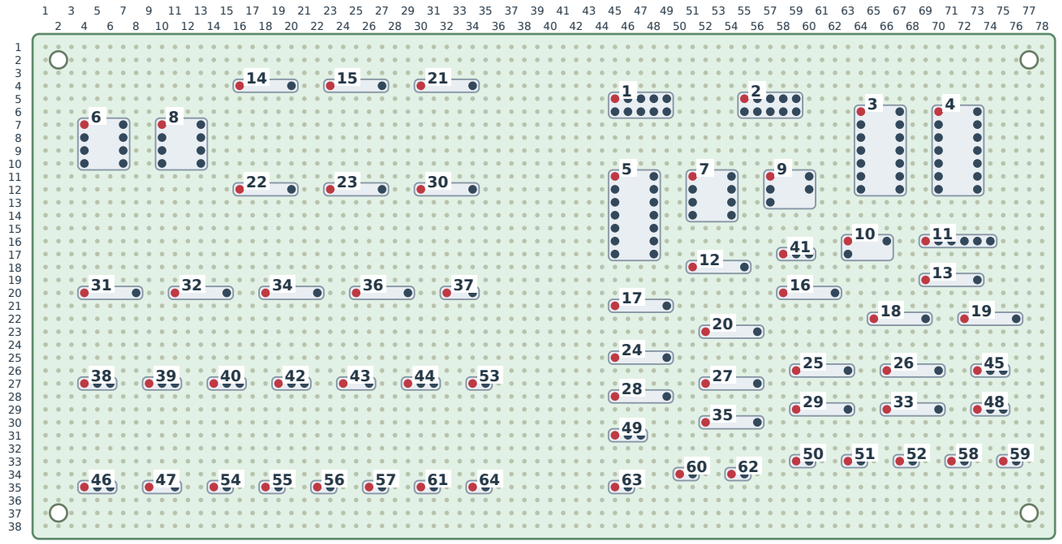


Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — placement

TOP coordinates retained on MIRRORED UNDERSIDE. Numbered reference tables follow.

power — TOP

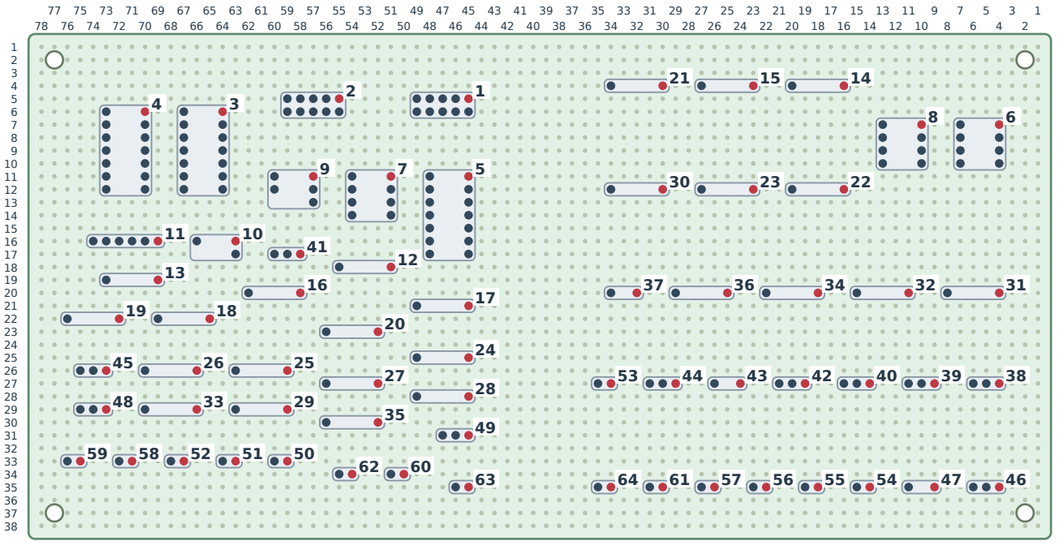


200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

power — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

power — components and endpoints

Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
1	J_POWER_INTERLOCK_MATRIX: POWER_MATRIX v2	CP09	1=45,5; 2=45,6; 3=46,5; 4=46,6; 5=47,5; 6=47,6; 7=48,5; 8=48,6; 9=49,5; 10=49,6	
2	J_POWER_INTERLOCK_SMART: POWER_SMART v2	CP09	1=55,5; 2=55,6; 3=56,5; 4=56,6; 5=57,5; 6=57,6; 7=58,5; 8=58,6; 9=59,5; 10=59,6	
3	U_POWER_INTERLOCK_FAULT: SN74HC74N	CP11	1=64,6; 2=64,7; 3=64,8; 4=64,9; 5=64,10; 6=64,11; 7=64,12; 8=67,12; 9=67,11; 10=67,10; 11=67,9; 12=67,8; 13=67,7; 14=67,6	
4	U_POWER_INTERLOCK_ALLOW: SN74HC08N	CP11	1=70,6; 2=70,7; 3=70,8; 4=70,9; 5=70,10; 6=70,11; 7=70,12; 8=73,12; 9=73,11; 10=73,10; 11=73,9; 12=73,8; 13=73,7; 14=73,6	
5	U_POWER_INTERLOCK_RESET: SN74HC00N	CP11	1=45,11; 2=45,12; 3=45,13; 4=45,14; 5=45,15; 6=45,16; 7=45,17; 8=48,17; 9=48,16; 10=48,15; 11=48,14; 12=48,13; 13=48,12; 14=48,11	
6	U_POWER_GUARDS_OV_5V: LTC4365ITS8#TRMPBF	CP10	1=4,7; 2=4,8; 3=4,9; 4=4,10; 5=7,10; 6=7,9; 7=7,8; 8=7,7	TSOT23-8 to numbered DIP8
7	U_POWER_ANALOG_TRIP: LM393P	CP11	1=51,11; 2=51,12; 3=51,13; 4=51,14; 5=54,14; 6=54,13; 7=54,12; 8=54,11	
8	U_POWER_GUARDS_OV_3V3: LTC4365ITS8#TRMPBF	CP10	1=10,7; 2=10,8; 3=10,9; 4=10,10; 5=13,10; 6=13,9; 7=13,8; 8=13,7	TSOT23-8 to numbered DIP8
9	U_POWER_ANALOG_SENSE: INA180A1IDBVR	CP11	1=57,11; 2=57,12; 3=57,13; 4=60,12; 5=60,11	SOT23-5 to numbered single-row adapter
10	U_POWER_INTERLOCK_POR: MCP130-300DI/TO	CP11	1=63,16; 2=63,17; 3=66,16	
11	J_POWER_INTERLOCK_TEST: scope / known load fixture	CP09	1=69,16; 2=70,16; 3=71,16; 4=72,16; 5=73,16; 6=74,16	
12	R_POWER_ANALOG_EDU_TOP: 47k	CP11	1=51,18; 2=55,18	
13	R_POWER_ANALOG_EDU_ON: 10k	CP11	1=69,19; 2=73,19	
14	R_POWER_GUARDS_OV_3V3_BOTTOM: 10k	CP10	1=16,4; 2=20,4	
15	R_POWER_GUARDS_UV_3V3_BOTTOM: 10k	CP10	1=23,4; 2=27,4	
16	R_POWER_INTERLOCK_EDU_FAULT_RESET_OFF: 100k	CP11	1=58,20; 2=62,20	
17	R_POWER_ANALOG_TRIP_TOP: 5.11k	CP11	1=45,21; 2=49,21	
18	R_POWER_ANALOG_TRIP_BOTTOM: 46.4k	CP11	1=65,22; 2=69,22	
19	R_POWER_ANALOG_EDU_BOTTOM: 47k	CP11	1=72,22; 2=76,22	
20	R_POWER_ANALOG_PRESET: 10k	CP11	1=52,23; 2=56,23	
21	R_POWER_GUARDS_OV_3V3_TOP: 59k	CP10	1=30,4; 2=34,4	
22	R_POWER_GUARDS_UV_3V3_TOP: 47.5k	CP10	1=16,12; 2=20,12	
23	D_POWER_INPUT_REVERSE: 1N5822	CP09	1=23,12; 2=27,12	
24	R_POWER_INTERLOCK_EDU_PWR_EN_OFF: 100k	CP11	1=45,25; 2=49,25	

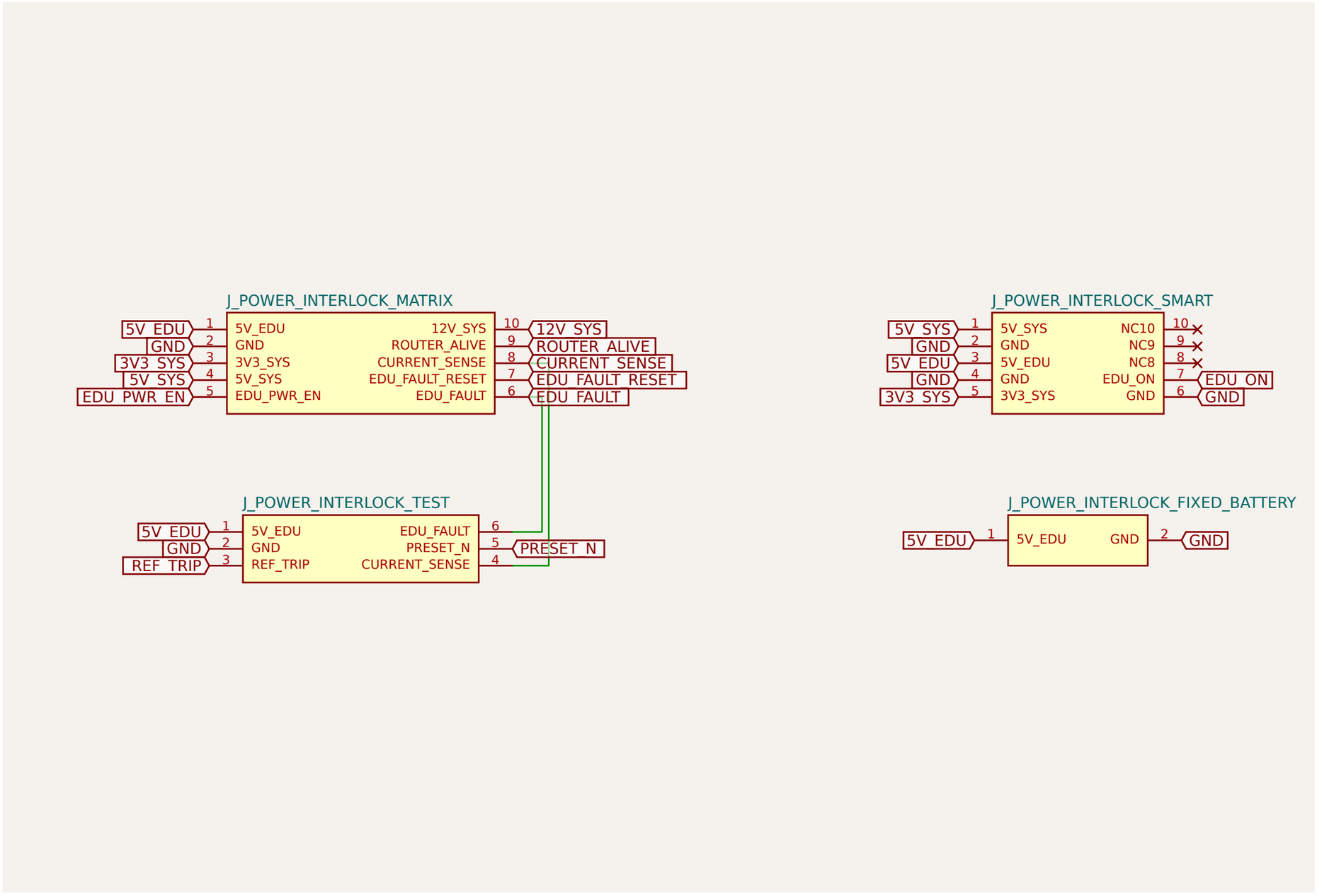
power — components and endpoints

Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
25	R_POWER_INTERLOCK_ROUTER_ALIVE_OFF: 100k	CP11	1=59,26; 2=63,26	
26	R_POWER_INTERLOCK_BATTERY_LED: 2.2k	CP11	1=66,26; 2=70,26	
27	R_POWER_INTERLOCK_LED_GATE_OFF: 100k	CP11	1=52,27; 2=56,27	
28	R_POWER_ANALOG_REF_FEED: 1k	CP11	1=45,28; 2=49,28	
29	R_POWER_ANALOG_GATE_OFF: 10k	CP11	1=59,29; 2=63,29	
30	R_POWER_GUARDS_OV_5V_BOTTOM: 10k	CP10	1=30,12; 2=34,12	
31	R_POWER_GUARDS_OV_5V_TOP: 92.8k	CP10	1=4,20; 2=8,20	
32	R_POWER_GUARDS_UV_5V_BOTTOM: 10k	CP10	1=11,20; 2=15,20	
33	F_POWER_ANALOG_EDU: MF-R050	CP11	1=66,29; 2=70,29	
34	R_POWER_GUARDS_UV_5V_TOP: 80.6k	CP10	1=18,20; 2=22,20	
35	R_POWER_ANALOG_SHUNT: 10	CP11	1=52,30; 2=56,30	
36	F_POWER_INPUT_INPUT: 2A slow fuse	CP09	1=25,20; 2=29,20	
37	C_POWER_GUARDS_OV_3V3_OUT: 470uF	CP10	1=32,20; 2=34,20	
38	J_POWER_INPUT_BUCK_3V3: External XL4015 set to 3.3V	CP09	1=4,27; 2=5,27; 3=6,27	Purchased XL4015 adjustable module
39	J_POWER_INPUT_BUCK_5V: External LM2596 set to 5V	CP09	1=9,27; 2=10,27; 3=11,27	Purchased LM2596 adjustable module
40	Q_POWER_GUARDS_OV_3V3_IN: IRLML6344TRPBF	CP10	1=14,27; 2=15,27; 3=16,27	SOT23 to numbered 3-pin adapter
41	Q_POWER_INTERLOCK_BATTERY_LED: 2N7000	CP11	1=58,17; 2=59,17; 3=60,17	
42	Q_POWER_GUARDS_OV_3V3_OUT: IRLML6344TRPBF	CP10	1=19,27; 2=20,27; 3=21,27	SOT23 to numbered 3-pin adapter
43	C_POWER_GUARDS_OV_5V_OUT: 470uF	CP10	1=24,27; 2=26,27	
44	Q_POWER_GUARDS_OV_5V_IN: IRLML6344TRPBF	CP10	1=29,27; 2=30,27; 3=31,27	SOT23 to numbered 3-pin adapter
45	U_POWER_ANALOG_REF: TL431ACLP	CP11	1=73,26; 2=74,26; 3=75,26	
46	Q_POWER_GUARDS_OV_5V_OUT: IRLML6344TRPBF	CP10	1=4,35; 2=5,35; 3=6,35	SOT23 to numbered 3-pin adapter
47	C_POWER_INPUT_INPUT: 470uF	CP09	1=9,35; 2=11,35	
48	Q_POWER_ANALOG_PASS: NDP6020P	CP11	1=73,29; 2=74,29; 3=75,29	

power — components and endpoints

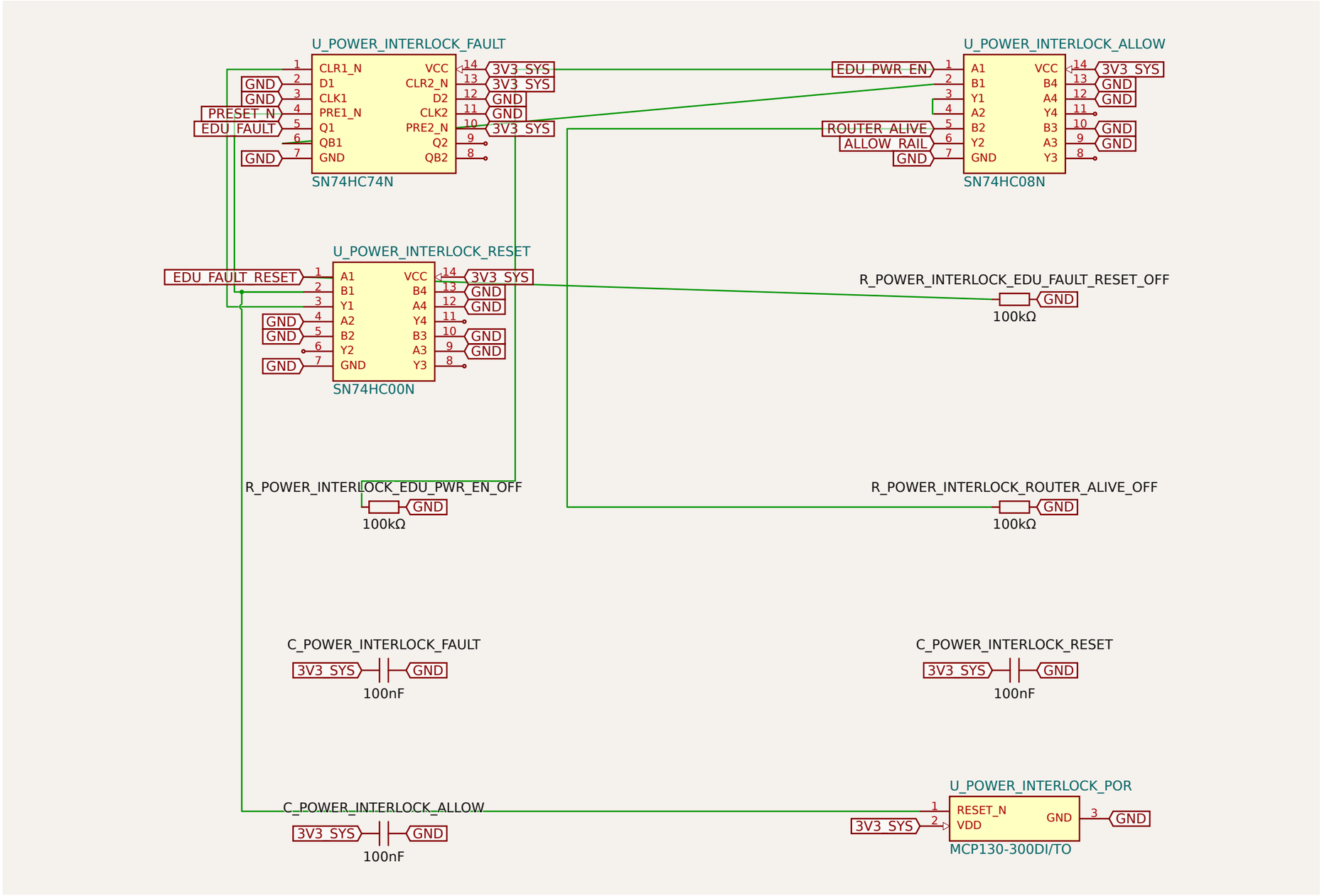
Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
49	Q_POWER_ANALOG_ENABLE: 2N7000	CP11	1=45,31; 2=46,31; 3=47,31	
50	C_POWER_ANALOG_TRIP: 100nF	CP11	1=59,33; 2=60,33	
51	LED_POWER_INTERLOCK_BATTERY: red LED — actual student rail on	CP11	1=63,33; 2=64,33	
52	J_POWER_INTERLOCK_FIXED_BATTERY: fixed battery patch lead: C2 positive, C1 ground; unplug before matrix mode	CP09	1=67,33; 2=68,33	
53	C_POWER_GUARDS_OV_3V3_IN: 100nF	CP10	1=34,27; 2=35,27	
54	JP_POWER_INPUT_LOGIC: removable 5-V consumer link	CP09	1=14,35; 2=15,35	
55	C_POWER_INPUT_INPUT_HF: 100nF	CP09	1=18,35; 2=19,35	
56	JP_POWER_INPUT_3V3: removable 3.3-V consumer link	CP09	1=22,35; 2=23,35	
57	C_POWER_GUARDS_OV_5V_IN: 100nF	CP10	1=26,35; 2=27,35	
58	C_POWER_INTERLOCK_FAULT: 100nF	CP11	1=71,33; 2=72,33	
59	C_POWER_INTERLOCK_RESET: 100nF	CP11	1=75,33; 2=76,33	
60	C_POWER_ANALOG_SENSE: 100nF	CP11	1=50,34; 2=51,34	
61	J_POWER_INPUT_SW_MAIN: latching switch, 3A at 12V DC	CP09	1=30,35; 2=31,35	
62	C_POWER_INTERLOCK_ALLOW: 100nF	CP11	1=54,34; 2=55,34	
63	C_POWER_ANALOG_EDU: 100nF	CP11	1=45,35; 2=46,35	
64	J_POWER_INPUT_INPUT: 5.5x2.1 mm barrel pigtail, centre positive	CP09	1=34,35; 2=35,35	

power — power-interlock: Outputs



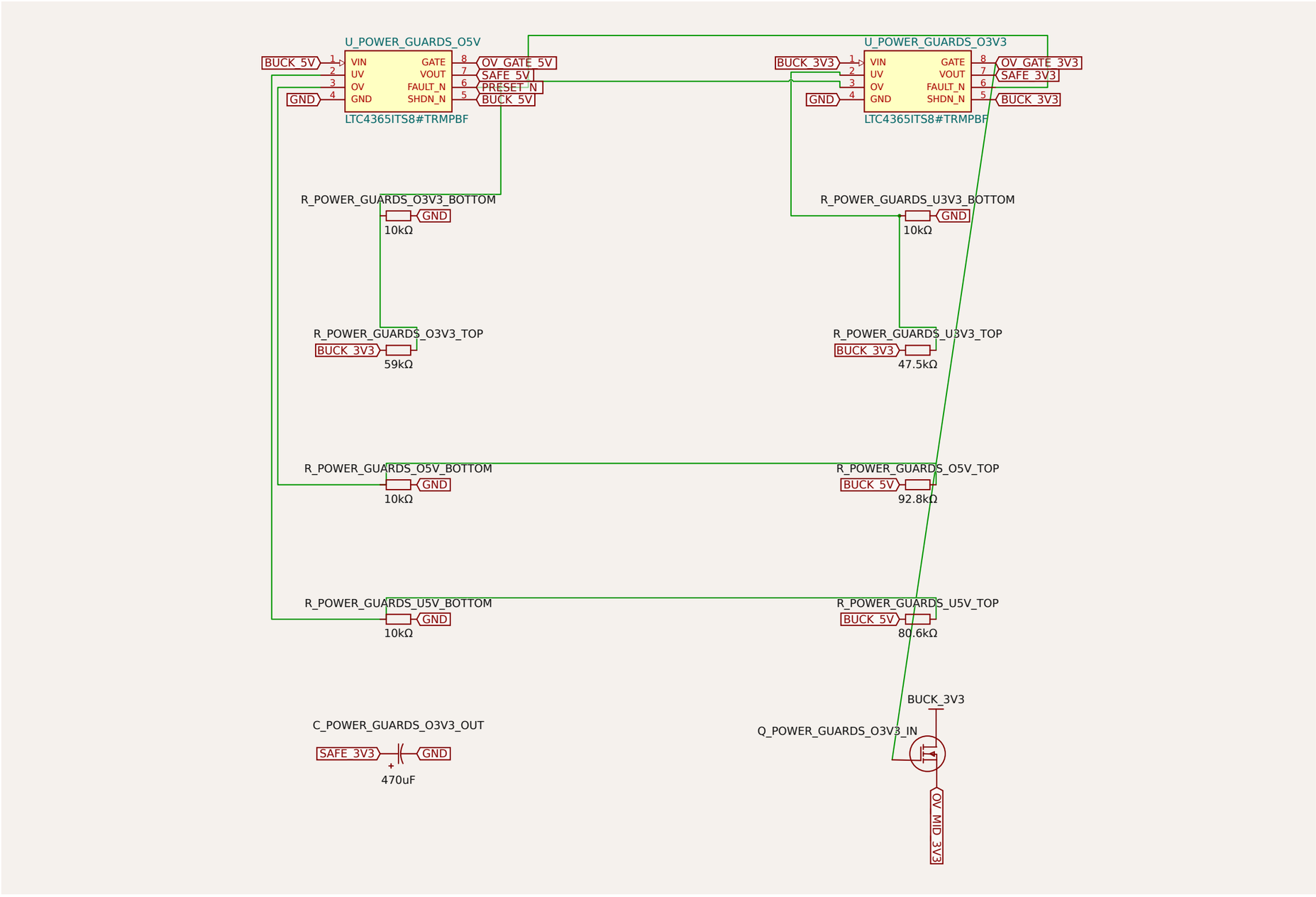
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-interlock: Latch



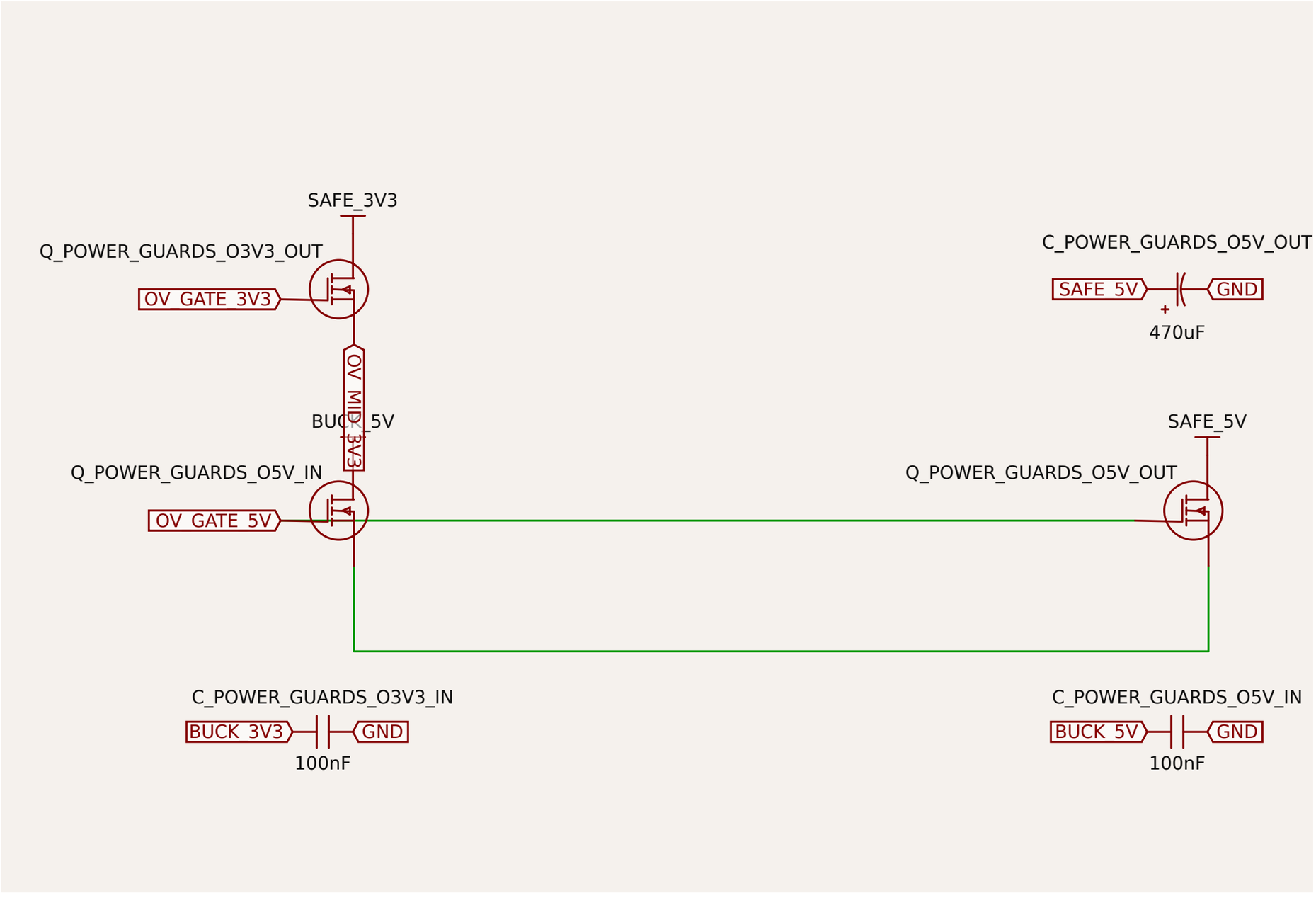
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-guards: VoltageGuard 1



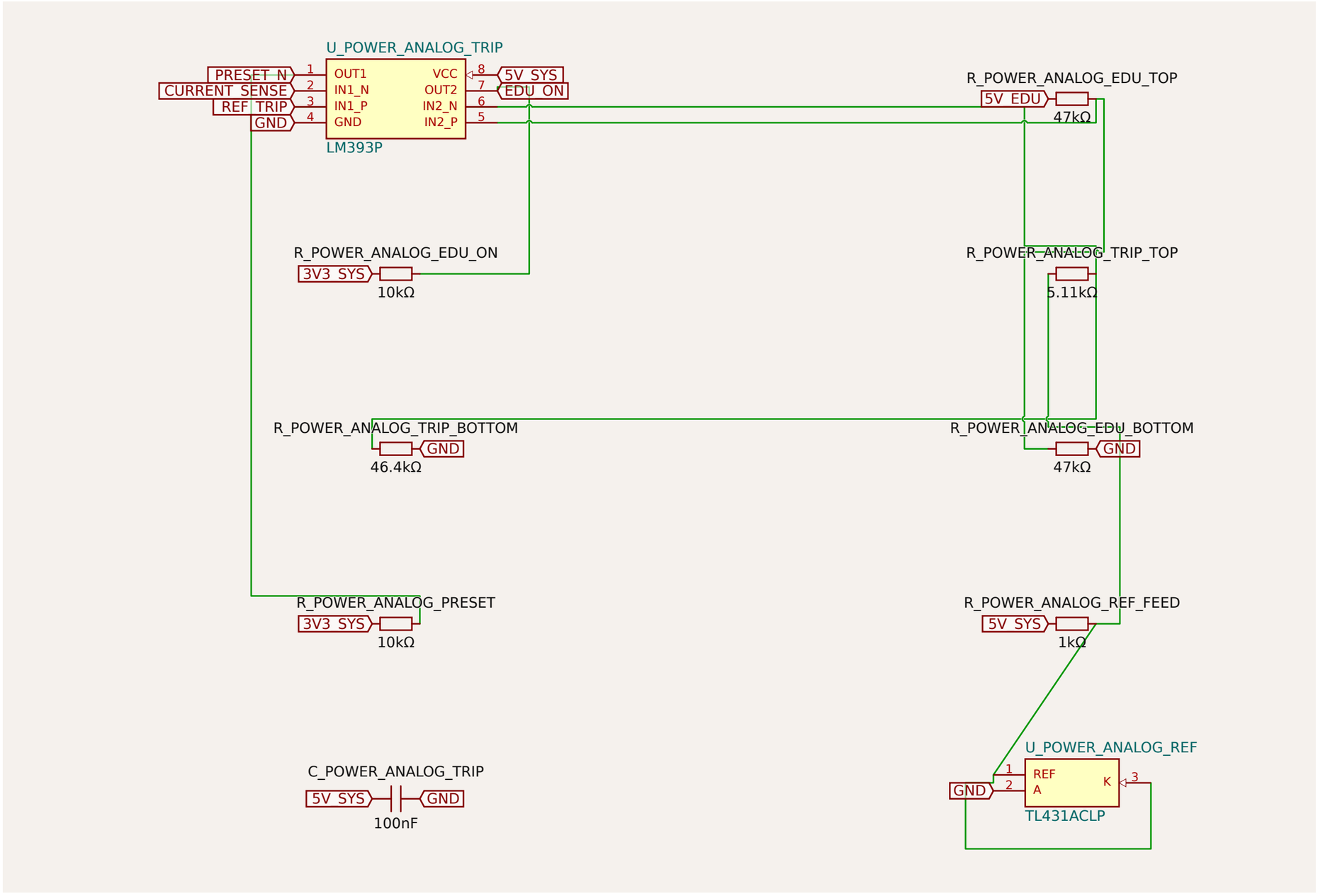
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-guards: VoltageGuard 2



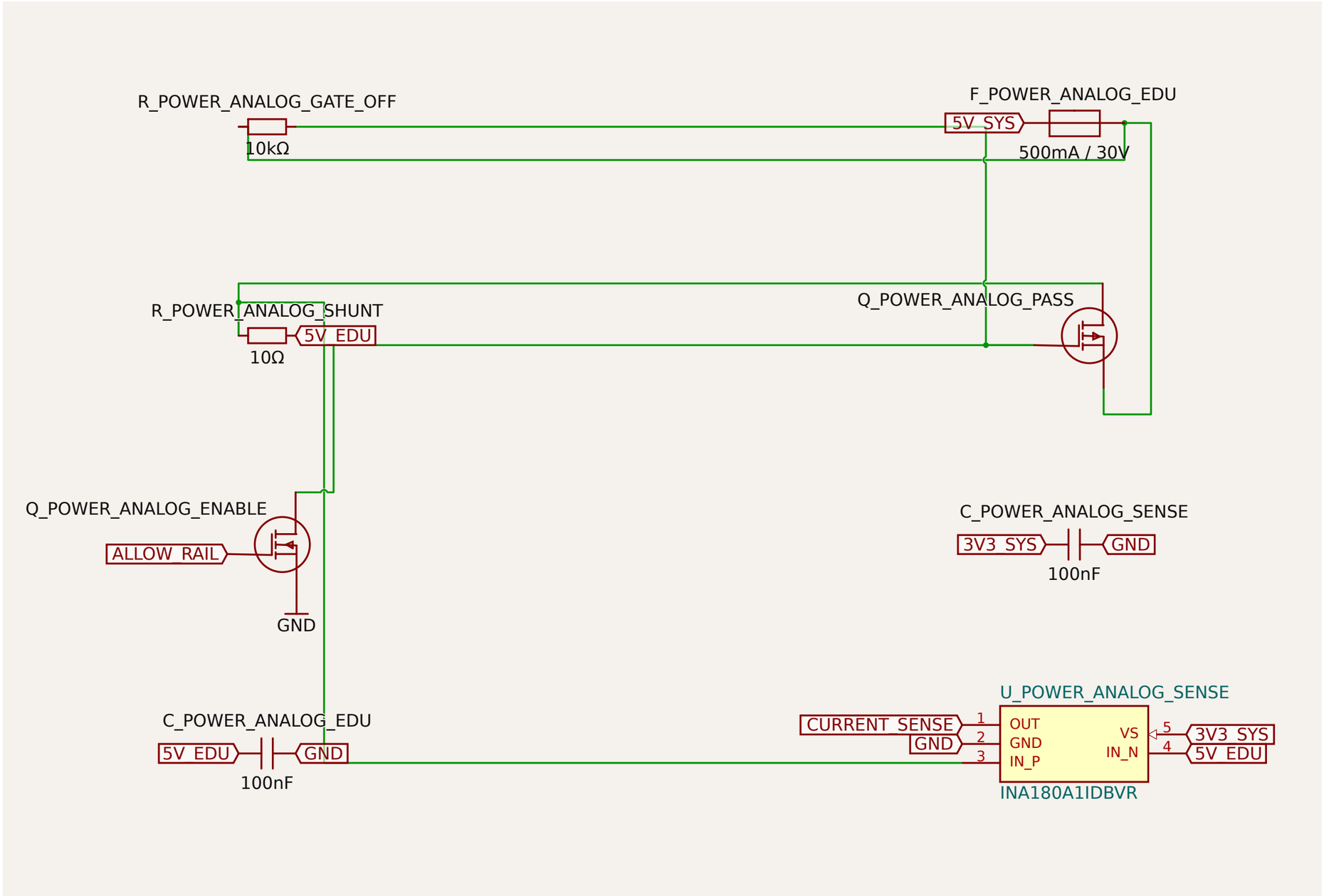
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-analog: Reference



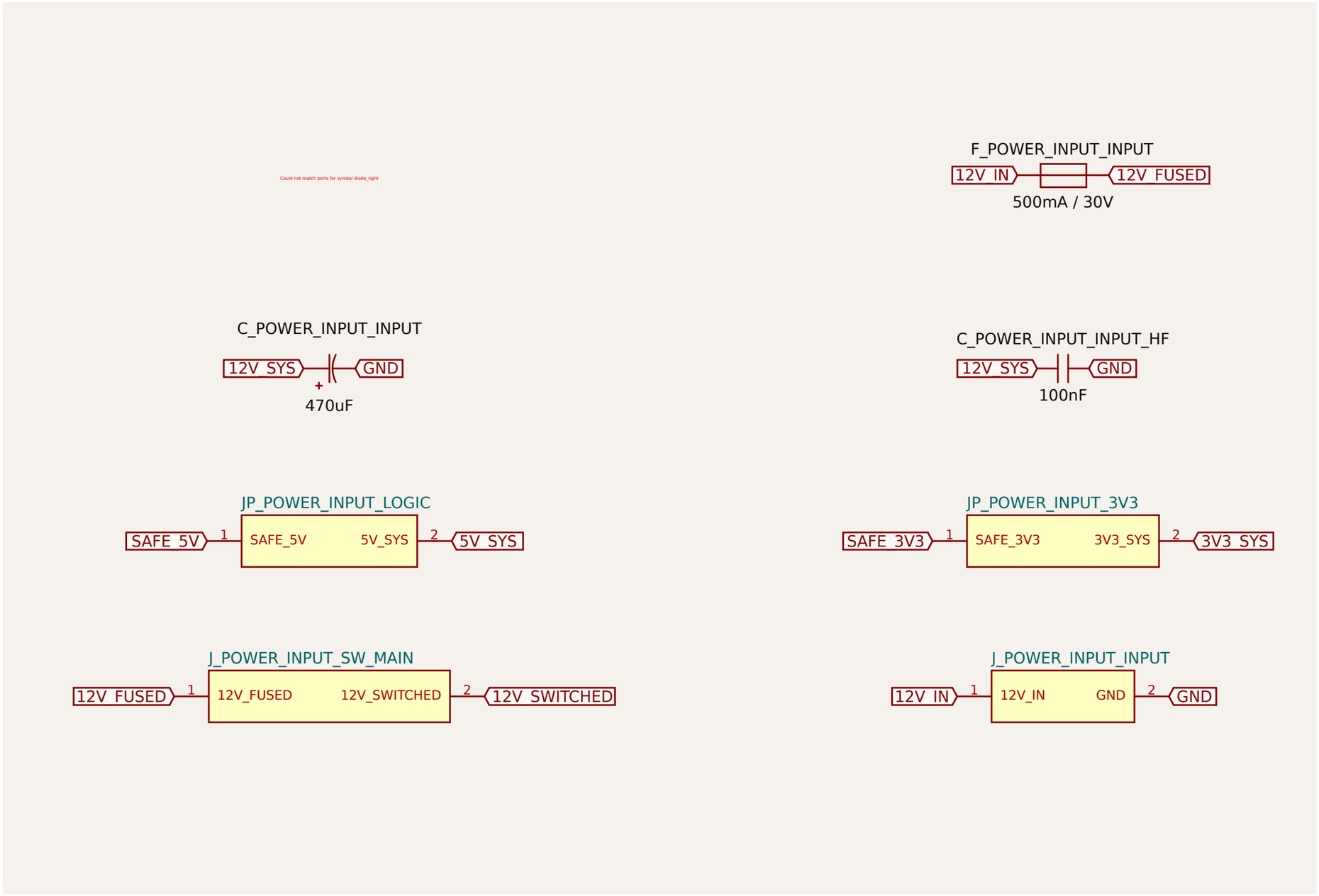
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-analog: Student



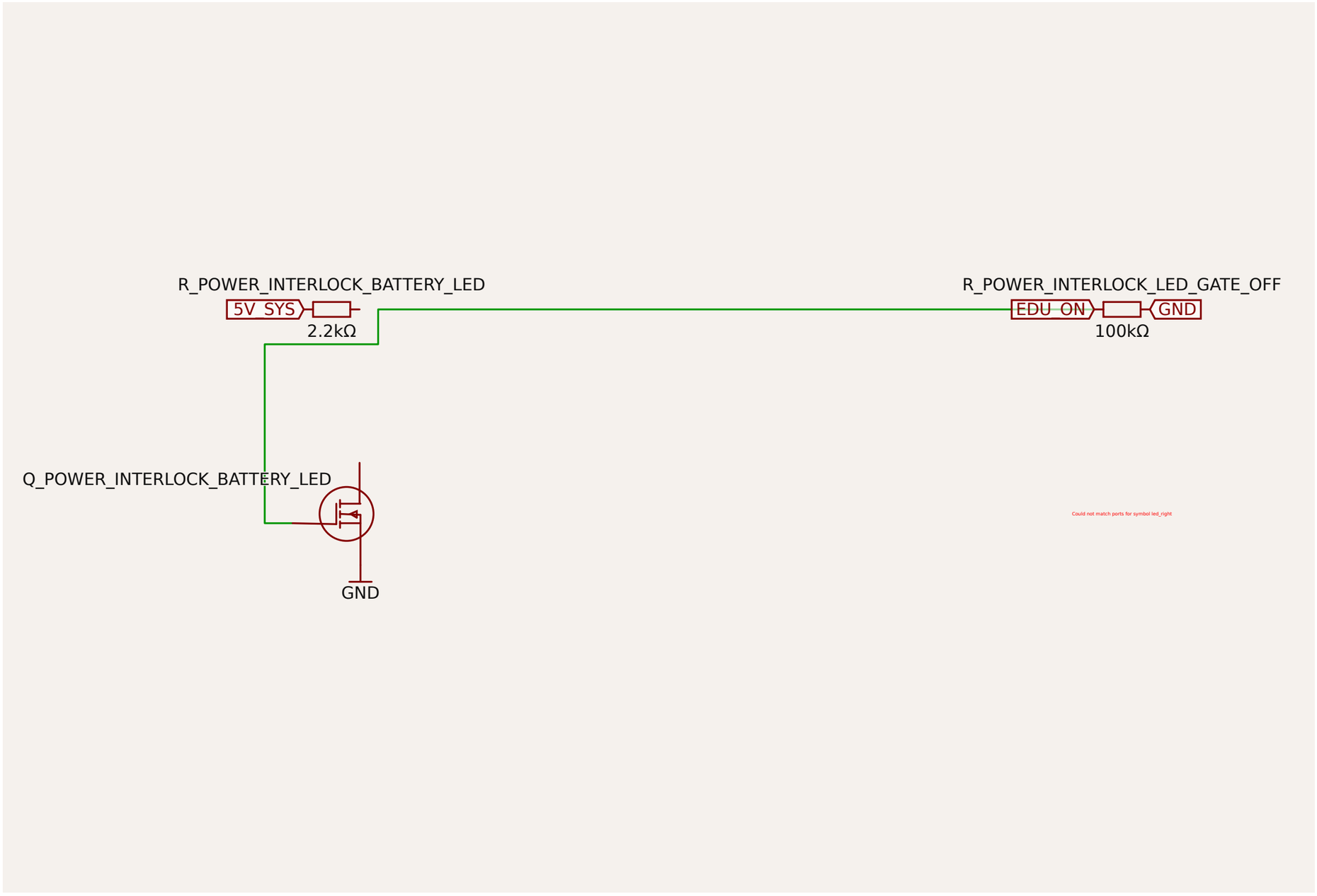
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-input: Input



Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-interlock: Indicator



Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

power — power-input: Circuit

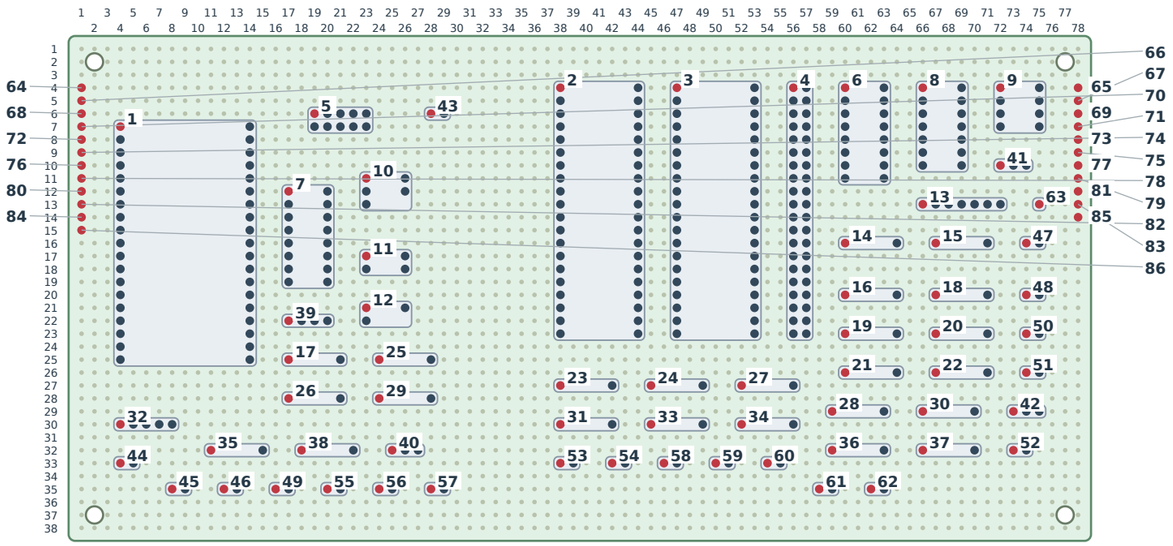


Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — placement

TOP coordinates retained on MIRRORED UNDERSIDE. Numbered reference tables follow.

router-matrix — TOP

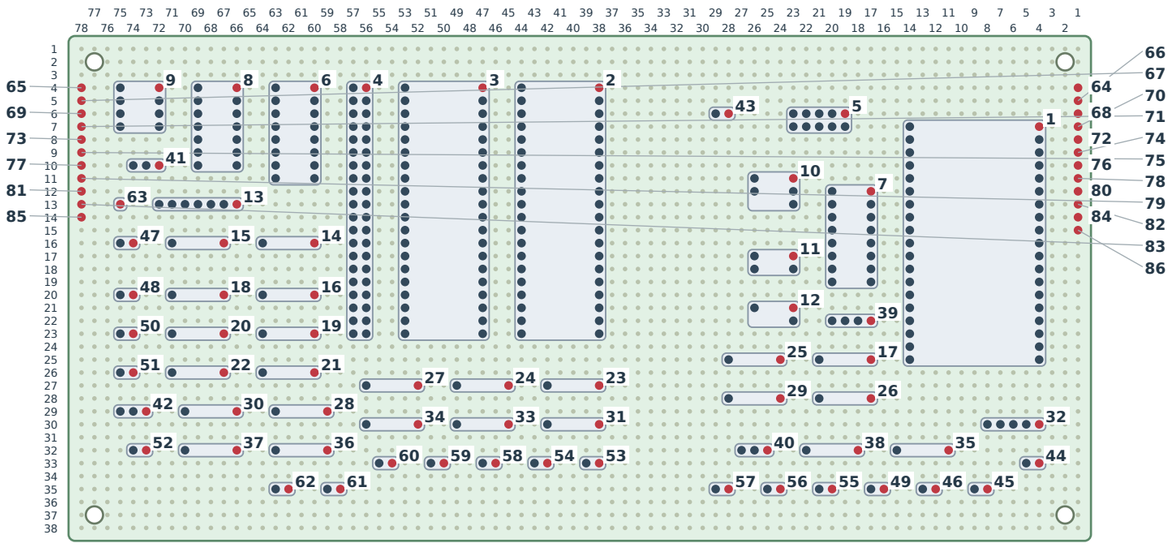


200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

router-matrix — MIRRORED UNDERSIDE



200×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered table.

PROTOTYPE: physical fit and hardware tests pending.

Not released for soldering. Software checks do not establish measured fit, manual-wire safety, protection response or PCB manufacturability.

router-matrix — components and endpoints

Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
1	U_ROUTER_CORE_ESP1: ESP32 DevKitC-38 — router	CP00	1=4,7; 2=4,8; 3=4,9; 4=4,10; 5=4,11; 6=4,12; 7=4,13; 8=4,14; 9=4,15; 10=4,16; 11=4,17; 12=4,18; 13=4,19; 14=4,20; 15=4,21; 16=4,22; 17=4,23; 18=4,24; 19=4,25; 20=14,7; 21=14,8; 22=14,9; 23=14,10; 24=14,11; 25=14,12; 26=14,13; 27=14,14; 28=14,15; 29=14,16; 30=14,17; 31=14,18; 32=14,19; 33=14,20; 34=14,21; 35=14,22; 36=14,23; 37=14,24; 38=14,25	
2	U_MATRIX_SWITCHES_MT_A: MT8816AE1	CP24	1=38,4; 2=38,5; 3=38,6; 4=38,7; 5=38,8; 6=38,9; 7=38,10; 8=38,11; 9=38,12; 10=38,13; 11=38,14; 12=38,15; 13=38,16; 14=38,17; 15=38,18; 16=38,19; 17=38,20; 18=38,21; 19=38,22; 20=38,23; 21=44,23; 22=44,22; 23=44,21; 24=44,20; 25=44,19; 26=44,18; 27=44,17; 28=44,16; 29=44,15; 30=44,14; 31=44,13; 32=44,12; 33=44,11; 34=44,10; 35=44,9; 36=44,8; 37=44,7; 38=44,6; 39=44,5; 40=44,4	
3	U_MATRIX_SWITCHES_MT_B: MT8816AE1	CP24	1=47,4; 2=47,5; 3=47,6; 4=47,7; 5=47,8; 6=47,9; 7=47,10; 8=47,11; 9=47,12; 10=47,13; 11=47,14; 12=47,15; 13=47,16; 14=47,17; 15=47,18; 16=47,19; 17=47,20; 18=47,21; 19=47,22; 20=47,23; 21=53,23; 22=53,22; 23=53,21; 24=53,20; 25=53,19; 26=53,18; 27=53,17; 28=53,16; 29=53,15; 30=53,14; 31=53,13; 32=53,12; 33=53,11; 34=53,10; 35=53,9; 36=53,8; 37=53,7; 38=53,6; 39=53,5; 40=53,4	
4	J_INTEGRATION_PI: Pi physical header	CP32	1=56,4; 2=57,4; 3=56,5; 4=57,5; 5=56,6; 6=57,6; 7=56,7; 8=57,7; 9=56,8; 10=57,8; 11=56,9; 12=57,9; 13=56,10; 14=57,10; 15=56,11; 16=57,11; 17=56,12; 18=57,12; 19=56,13; 20=57,13; 21=56,14; 22=57,14; 23=56,15; 24=57,15; 25=56,16; 26=57,16; 27=56,17; 28=57,17; 29=56,18; 30=57,18; 31=56,19; 32=57,19; 33=56,20; 34=57,20; 35=56,21; 36=57,21; 37=56,22; 38=57,22; 39=56,23; 40=57,23	
5	J_ROUTER_CORE_POWER: POWER_MATRIX v2	CP11	1=19,6; 2=19,7; 3=20,6; 4=20,7; 5=21,6; 6=21,7; 7=22,6; 8=22,7; 9=23,6; 10=23,7	
6	U_MATRIX_CONTROL_ADDR: SN74HCT595N	CP24	1=60,4; 2=60,5; 3=60,6; 4=60,7; 5=60,8; 6=60,9; 7=60,10; 8=60,11; 9=63,11; 10=63,10; 11=63,9; 12=63,8; 13=63,7; 14=63,6; 15=63,5; 16=63,4	
7	U_ROUTER_SAFETY_WATCH: SN74HC123N	CP13	1=17,12; 2=17,13; 3=17,14; 4=17,15; 5=17,16; 6=17,17; 7=17,18; 8=17,19; 9=20,19; 10=20,18; 11=20,17; 12=20,16; 13=20,15; 14=20,14; 15=20,13; 16=20,12	
8	U_MATRIX_CONTROL_BUFFER: SN74HCT125N	CP24	1=66,4; 2=66,5; 3=66,6; 4=66,7; 5=66,8; 6=66,9; 7=66,10; 8=69,10; 9=69,9; 10=69,8; 11=69,7; 12=69,6; 13=69,5; 14=69,4	
9	U_MATRIX_CONTROL_DIAG: OPA2333AID	CP24	1=72,4; 2=72,5; 3=72,6; 4=72,7; 5=75,7; 6=75,6; 7=75,5; 8=75,4	SOIC8 to DIP8
10	U_ROUTER_SAFETY_FAULT_RETURN: SN74LVC1G125DBVR	CP11	1=23,11; 2=23,12; 3=23,13; 4=26,12; 5=26,11	SOT23-5 to numbered adapter
11	K_ROUTER_SAFETY_ADC: AQY212EH	CP11	1=23,17; 2=23,18; 3=26,18; 4=26,17	
12	U_ROUTER_SAFETY_USB_POR: MCP130-300DI/TO	CP13	1=23,21; 2=23,22; 3=26,21	
13	J_INTEGRATION_RFID: RC522 labelled SPI carrier interface	CP34	1=66,13; 2=67,13; 3=68,13; 4=69,13; 5=70,13; 6=71,13; 7=72,13	Purchased RC522 reader
14	R_MATRIX_CONTROL_STB_B_IN_OFF: 100k	CP24	1=60,16; 2=64,16	
15	R_MATRIX_CONTROL_SOURCE_LO: 470	CP24	1=67,16; 2=71,16	
16	R_MATRIX_CONTROL_DIAG_TOP: 1M	CP24	1=60,20; 2=64,20	
17	R_ROUTER_SAFETY_BUF_OE: 10k	CP13	1=17,25; 2=21,25	
18	R_MATRIX_CONTROL_STB_A_IN_OFF: 100k	CP24	1=67,20; 2=71,20	
19	R_MATRIX_CONTROL_SOURCE_HI: 10k	CP24	1=60,23; 2=64,23	
20	R_MATRIX_CONTROL_CAL: 1k	CP24	1=67,23; 2=71,23	
21	R_MATRIX_CONTROL_XBAR_LATCH_OFF: 100k	CP24	1=60,26; 2=64,26	
22	D_MATRIX_CONTROL_SOURCE: BAT85	CP24	1=67,26; 2=71,26	
23	R_MATRIX_CONTROL_DISCHARGE: 470	CP24	1=38,27; 2=42,27	
24	R_MATRIX_CONTROL_DIAG_BOTTOM: 1M	CP24	1=45,27; 2=49,27	

router-matrix — components and endpoints

Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
25	R_ROUTER_SAFETY_ALIVE_OFF: 100k	CP13	1=24,25; 2=28,25	
26	R_ROUTER_SAFETY_KADC_LED: 330	CP11	1=17,28; 2=21,28	
27	R_MATRIX_CONTROL_XBAR_CLK_OFF: 100k	CP24	1=52,27; 2=56,27	
28	R_MATRIX_CONTROL_XBAR_DAT_OFF: 100k	CP24	1=59,29; 2=63,29	
29	R_ROUTER_SAFETY_USB_POR: 10k	CP13	1=24,28; 2=28,28	
30	R_MATRIX_SWITCHES_RESET_ON: 47k	CP24	1=66,29; 2=70,29	
31	R_MATRIX_SWITCHES_RST_RELEASE_OFF: 100k	CP24	1=38,30; 2=42,30	
32	J_ROUTER_CORE_FIXTURE: manual continuity fixture; ESP1 GPIO4 SDA / GPIO16 SCL / GPIO25 stimulus	CP00	1=4,30; 2=5,30; 3=6,30; 4=7,30; 5=8,30	
33	R_INTEGRATION_BACK: 10k	CP32	1=45,30; 2=49,30	
34	R_MATRIX_SWITCHES_STBB_OFF: 10k	CP24	1=52,30; 2=56,30	
35	R_ROUTER_SAFETY_WATCH: 549kΩ ±1% watchdog timing — qualify before permanent installation	CP13	1=11,32; 2=15,32	
36	R_INTEGRATION_ENERGIZE: 10k	CP32	1=59,32; 2=63,32	
37	R_MATRIX_SWITCHES_STBA_OFF: 10k	CP24	1=66,32; 2=70,32	
38	R_ROUTER_SAFETY_HEART: 100k	CP13	1=18,32; 2=22,32	
39	J_ROUTER_CORE_PROTOTYPE: USB breadboard prototype; remove before final rail connection	CP00	1=17,22; 2=18,22; 3=19,22; 4=20,22	
40	Q_ROUTER_SAFETY_BUF: 2N7000	CP13	1=25,32; 2=26,32; 3=27,32	
41	Q_MATRIX_SWITCHES_RESET_RELEASE: 2N7000	CP24	1=72,10; 2=73,10; 3=74,10	
42	Q_MATRIX_SWITCHES_RESET_ALIVE: 2N7000	CP24	1=73,29; 2=74,29; 3=75,29	
43	J_ROUTER_CORE_LIGHT: 5V white lighting — never 5V_EDU	CP32	1=28,6; 2=29,6	
44	C_ROUTER_SAFETY_RETURN: 100nF	CP11	1=4,33; 2=5,33	
45	J_ROUTER_CORE_FAN1: 12V fan 1 — fixed-speed	CP32	1=8,35; 2=9,35	
46	J_ROUTER_CORE_FAN2: 12V fan 2 — fixed-speed	CP32	1=12,35; 2=13,35	
47	J_INTEGRATION_SW_ENERGIZE: Energizar dry-contact button	CP32	1=74,16; 2=75,16	
48	C_MATRIX_CONTROL_BUFFER: 100nF	CP24	1=74,20; 2=75,20	

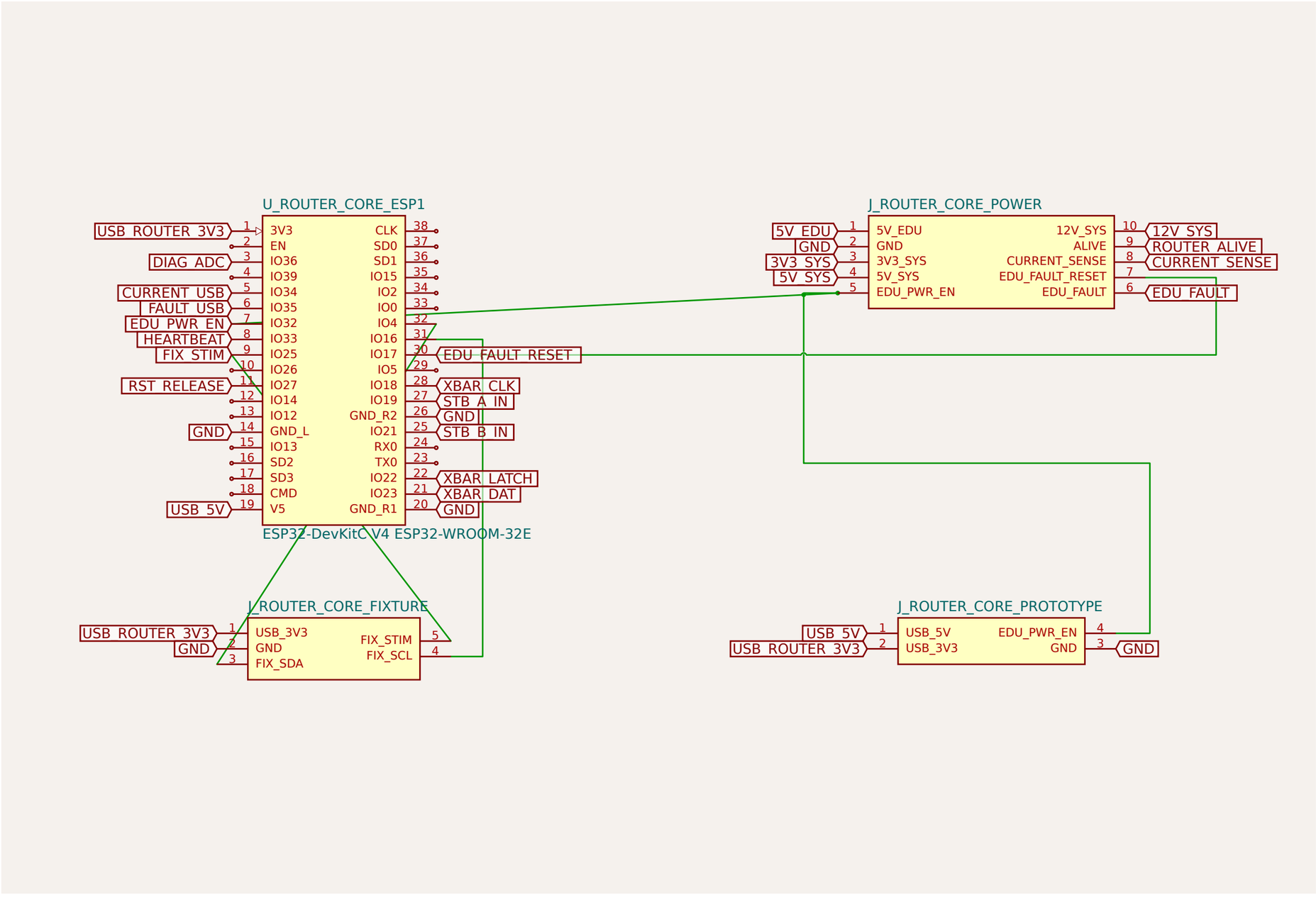
router-matrix — components and endpoints

Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
49	C_ROUTER_SAFETY_WATCH: 1µF ±5% film timing capacitor; not unverified inventory electrolytic	CP13	1=16,35; 2=17,35	
50	J_INTEGRATION_SW_BACK: Back dry-contact button	CP32	1=74,23; 2=75,23	
51	C_MATRIX_CONTROL_DIAG_VCC: 100nF	CP24	1=74,26; 2=75,26	
52	C_MATRIX_SWITCHES_MT0_BULK: 10uF	CP24	1=73,32; 2=74,32	
53	C_MATRIX_SWITCHES_MT1: 100nF	CP24	1=38,33; 2=39,33	
54	C_MATRIX_SWITCHES_MT1_BULK: 10uF	CP24	1=42,33; 2=43,33	
55	J_ROUTER_CORE_12V: Protected 12V input	CP32	1=20,35; 2=21,35	
56	J_ROUTER_CORE_5V: System 5V input	CP32	1=24,35; 2=25,35	
57	C_ROUTER_SAFETY_WATCH_VCC: 100nF	CP13	1=28,35; 2=29,35	
58	C_MATRIX_CONTROL_ADDR: 100nF	CP24	1=46,33; 2=47,33	
59	C_INTEGRATION_ENERGIZE: 100nF	CP32	1=50,33; 2=51,33	
60	C_INTEGRATION_BACK: 100nF	CP32	1=54,33; 2=55,33	
61	C_MATRIX_CONTROL_DIAG: 1nF	CP24	1=58,35; 2=59,35	
62	C_MATRIX_SWITCHES_MT0: 100nF	CP24	1=62,35; 2=63,35	
63	J_INTEGRATION_GROUND: ground to electronics	CP32	1=75,13	
64	J_BOUNDARY_1: Boundary: NODE00	CP24	1=1,4	
65	J_BOUNDARY_2: Boundary: NODE01	CP24	1=78,4	
66	J_BOUNDARY_3: Boundary: NODE02	CP24	1=1,5	
67	J_BOUNDARY_4: Boundary: NODE03	CP24	1=78,5	
68	J_BOUNDARY_5: Boundary: NODE04	CP24	1=1,6	
69	J_BOUNDARY_6: Boundary: NODE05	CP24	1=78,6	
70	J_BOUNDARY_7: Boundary: NODE06	CP24	1=1,7	
71	J_BOUNDARY_8: Boundary: NODE07	CP24	1=78,7	
72	J_BOUNDARY_9: Boundary: NODE08	CP24	1=1,8	

router-matrix — components and endpoints

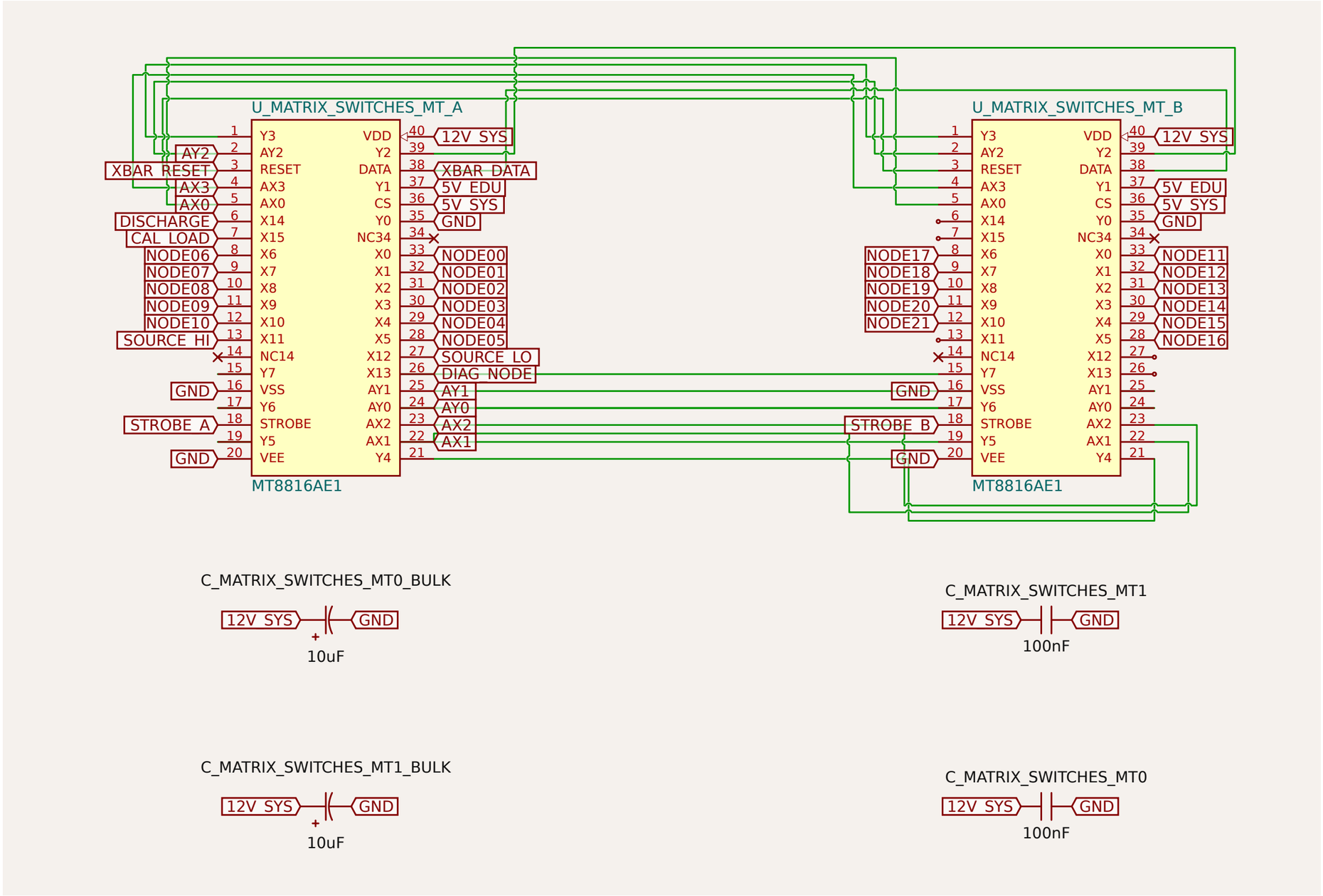
Badge	Ref / purpose	Install	TOP hole coordinates	Adapter / module
73	J_BOUNDARY_10: Boundary: NODE09	CP24	1=78,8	
74	J_BOUNDARY_11: Boundary: NODE10	CP24	1=1,9	
75	J_BOUNDARY_12: Boundary: NODE11	CP24	1=78,9	
76	J_BOUNDARY_13: Boundary: NODE12	CP24	1=1,10	
77	J_BOUNDARY_14: Boundary: NODE13	CP24	1=78,10	
78	J_BOUNDARY_15: Boundary: NODE14	CP24	1=1,11	
79	J_BOUNDARY_16: Boundary: NODE15	CP24	1=78,11	
80	J_BOUNDARY_17: Boundary: NODE16	CP24	1=1,12	
81	J_BOUNDARY_18: Boundary: NODE17	CP24	1=78,12	
82	J_BOUNDARY_19: Boundary: NODE18	CP24	1=1,13	
83	J_BOUNDARY_20: Boundary: NODE19	CP24	1=78,13	
84	J_BOUNDARY_21: Boundary: NODE20	CP24	1=1,14	
85	J_BOUNDARY_22: Boundary: NODE21	CP24	1=78,14	
86	J_BOUNDARY_23: Boundary: EDU_FAULT_RESET	CP00	1=1,15	

router-matrix — router-core: Controller



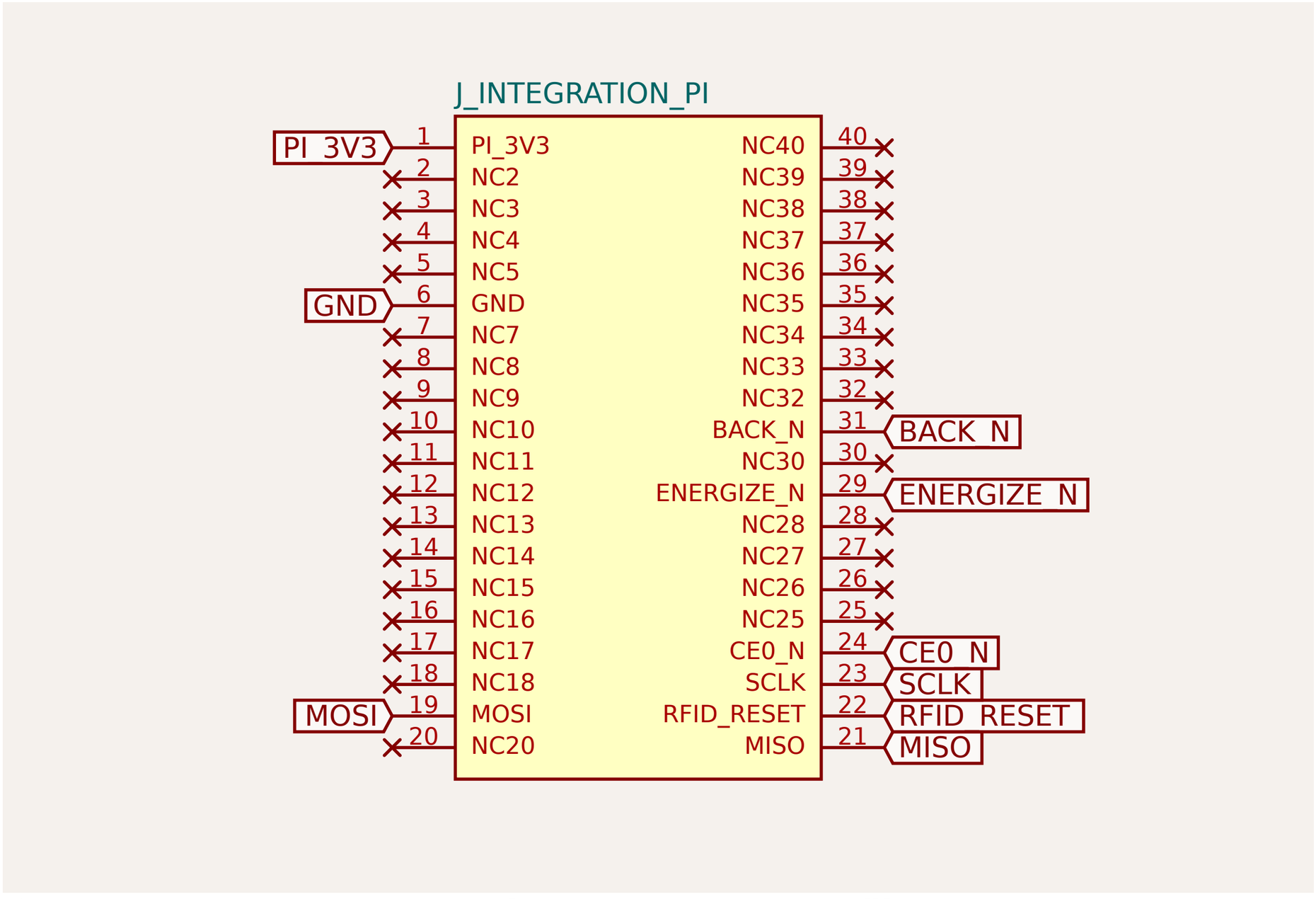
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — matrix-switches: Matrix



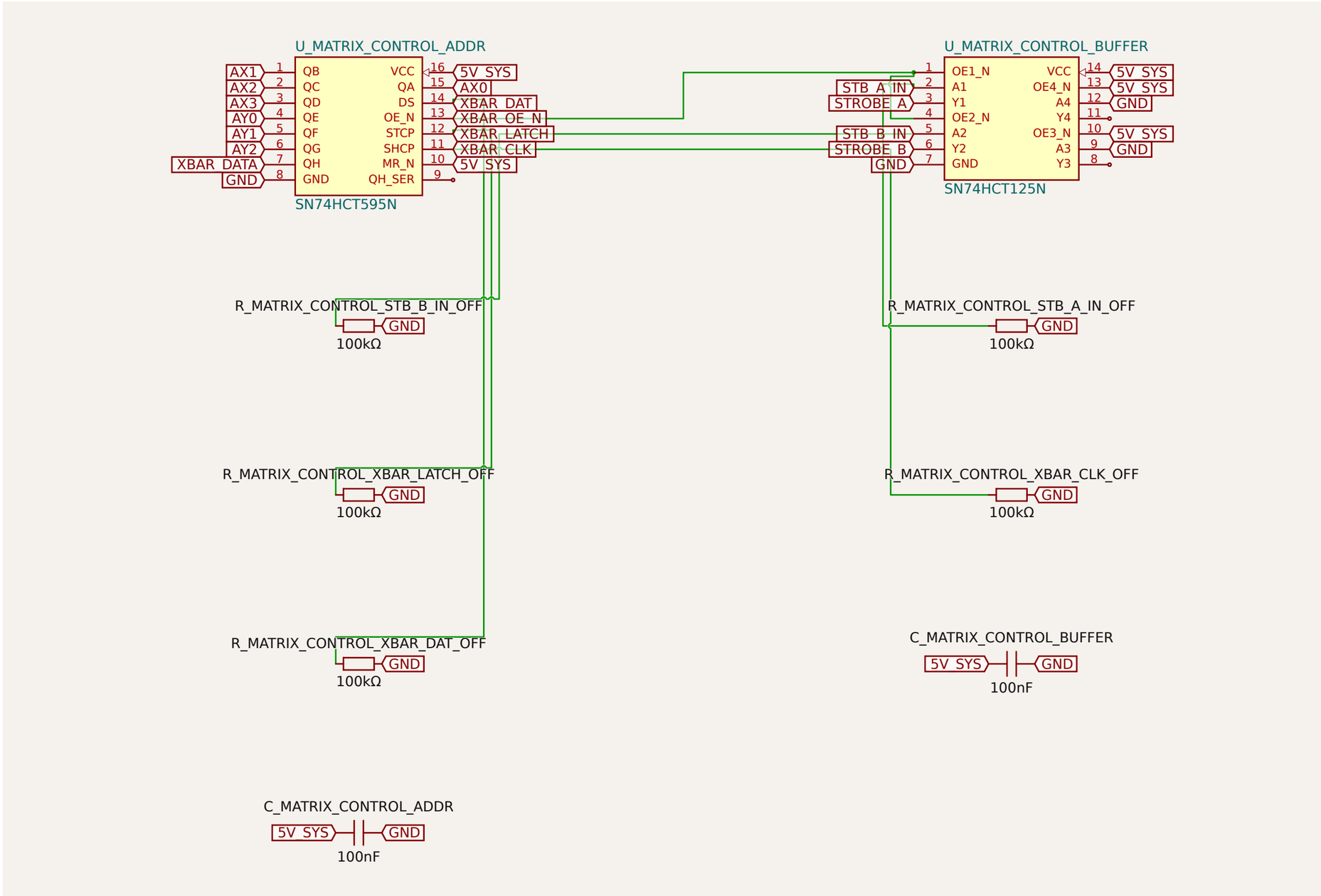
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — integration: Pi



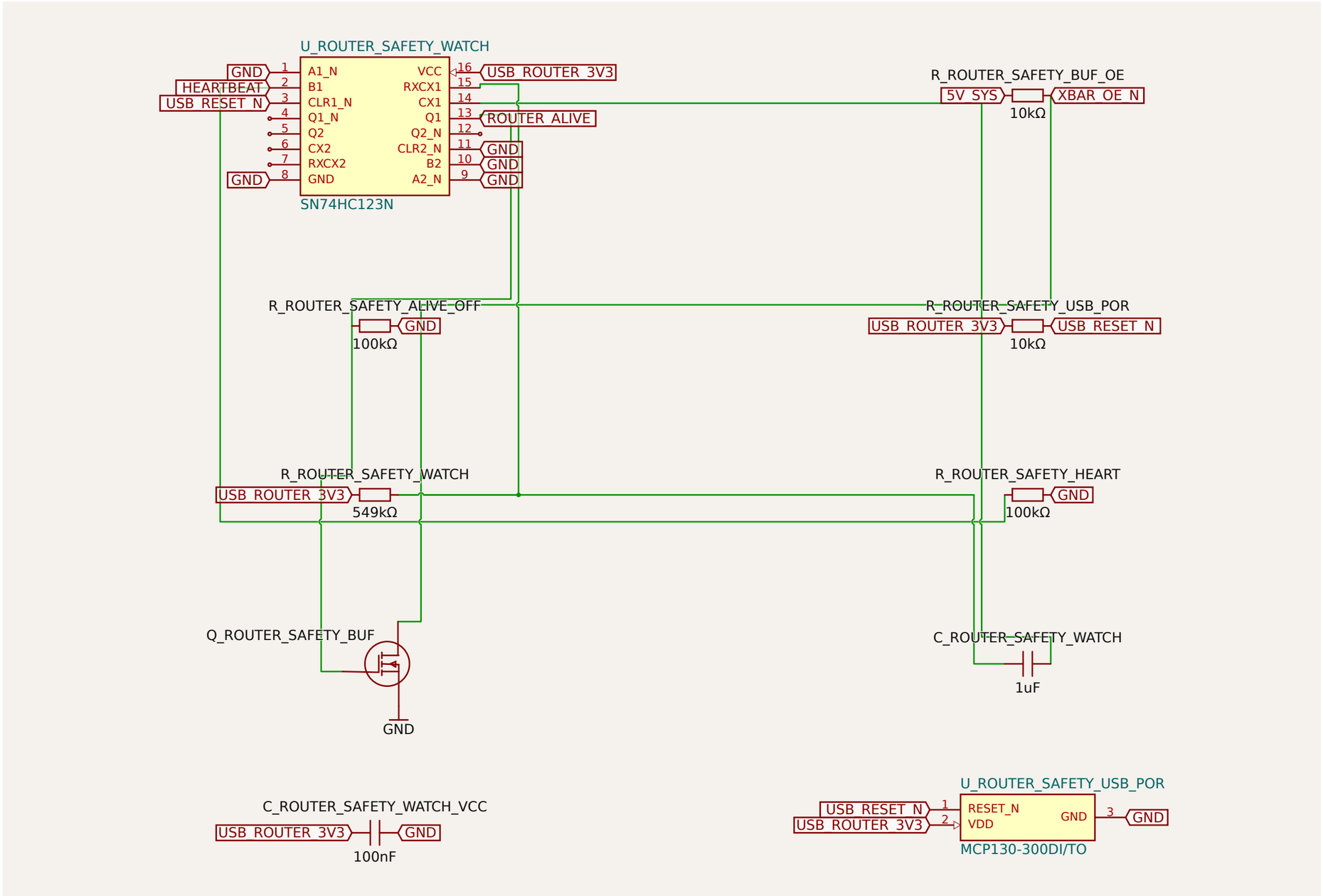
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — matrix-control: Control



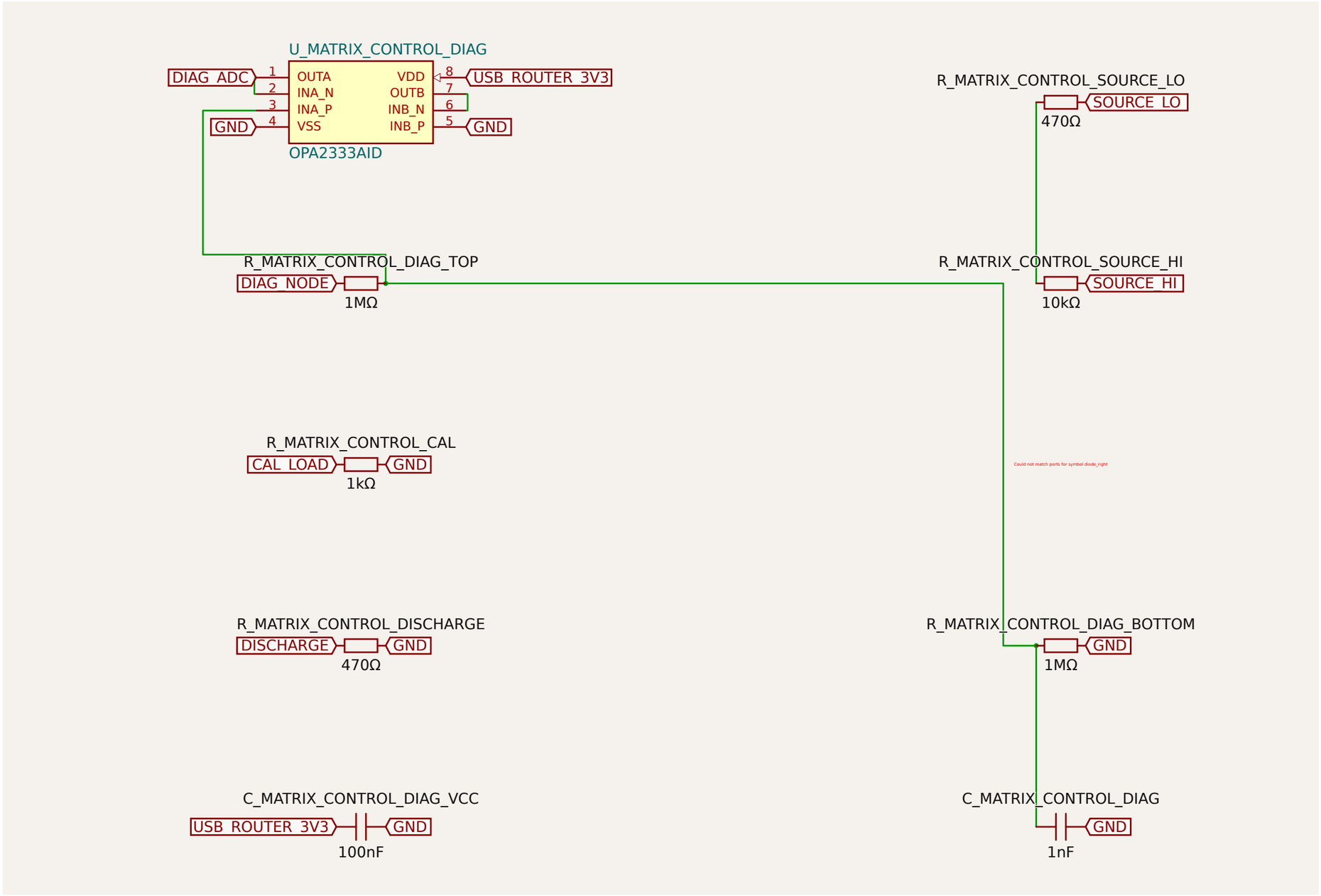
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — router-safety: Watchdog



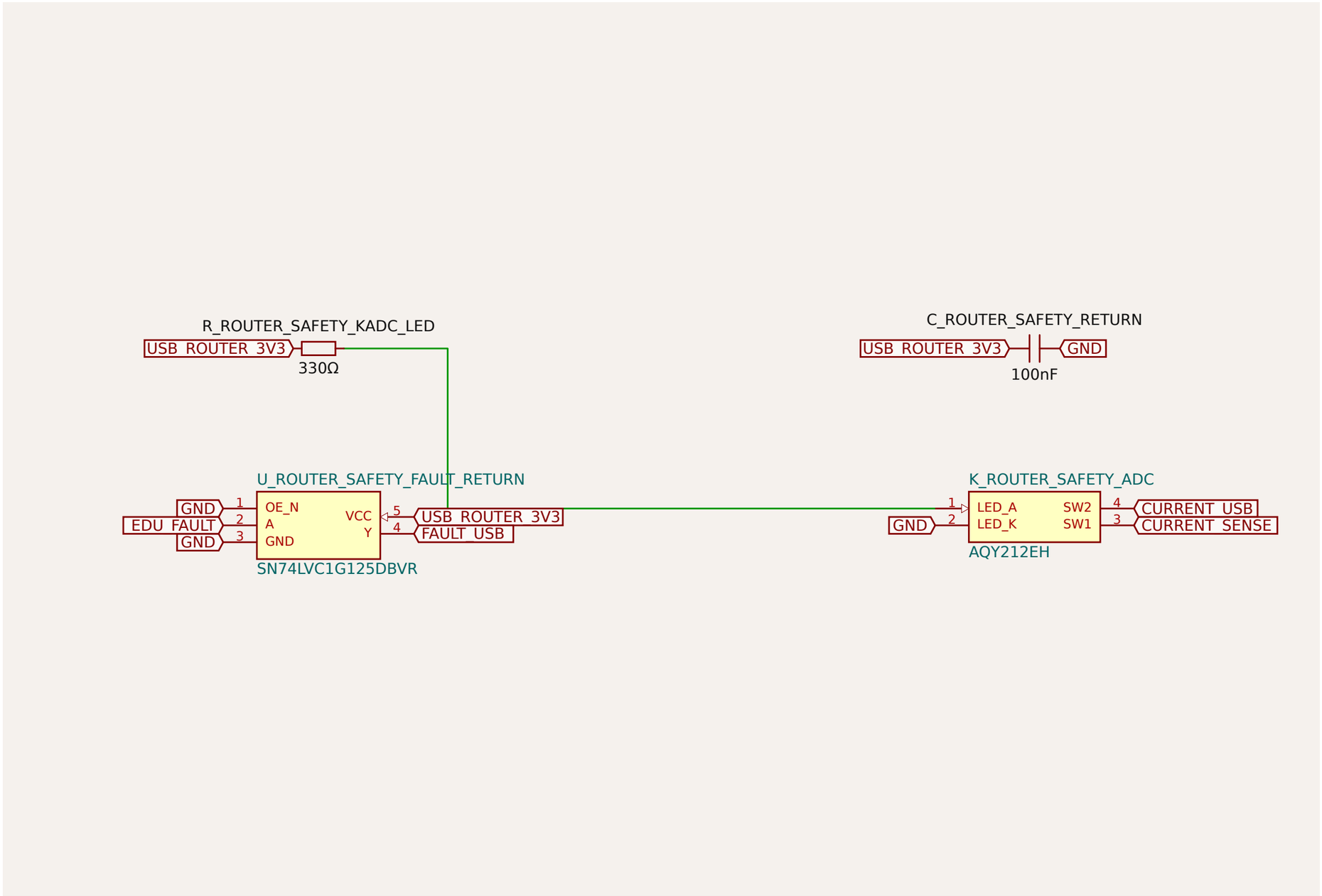
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — matrix-control: Tools



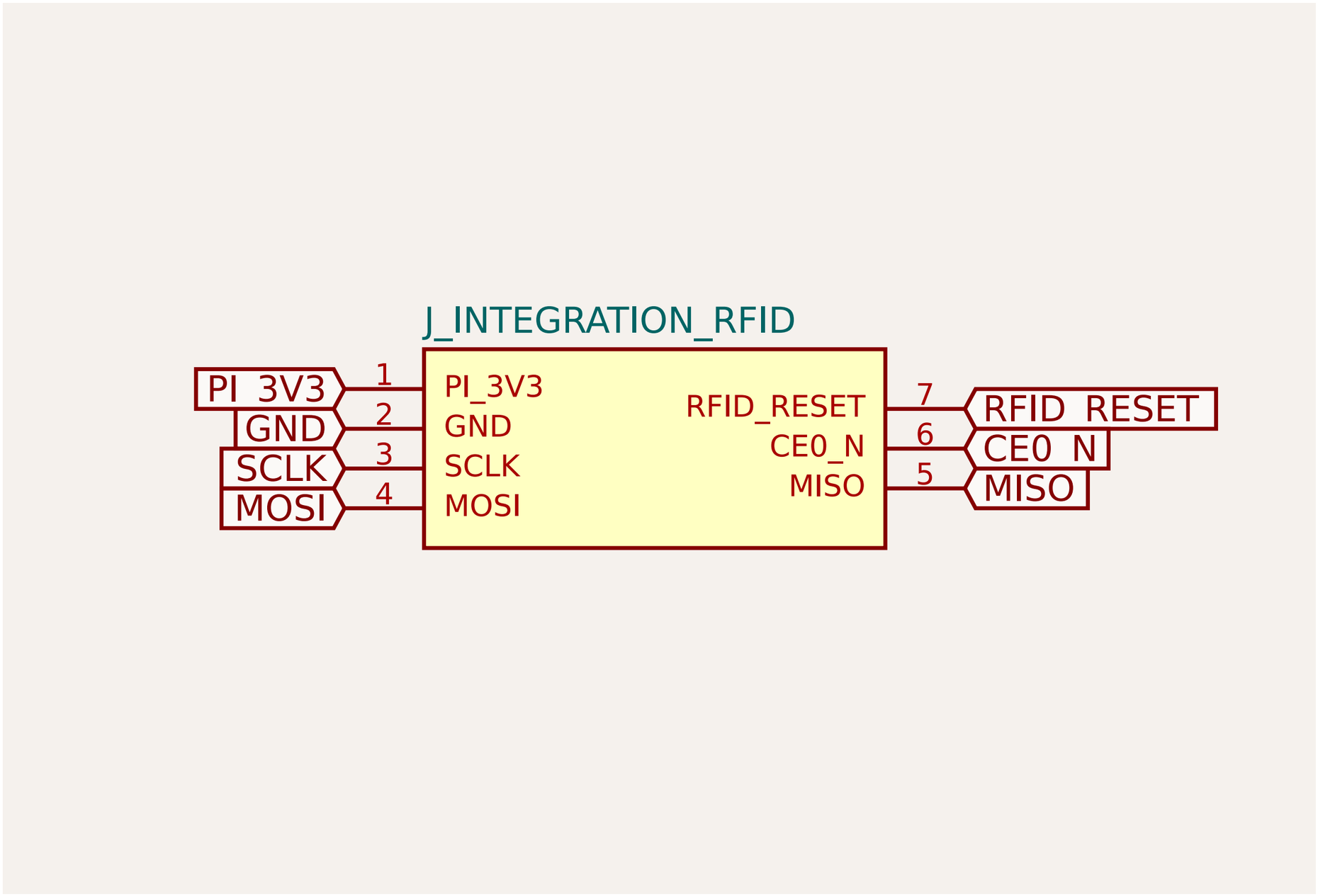
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — router-safety: Return



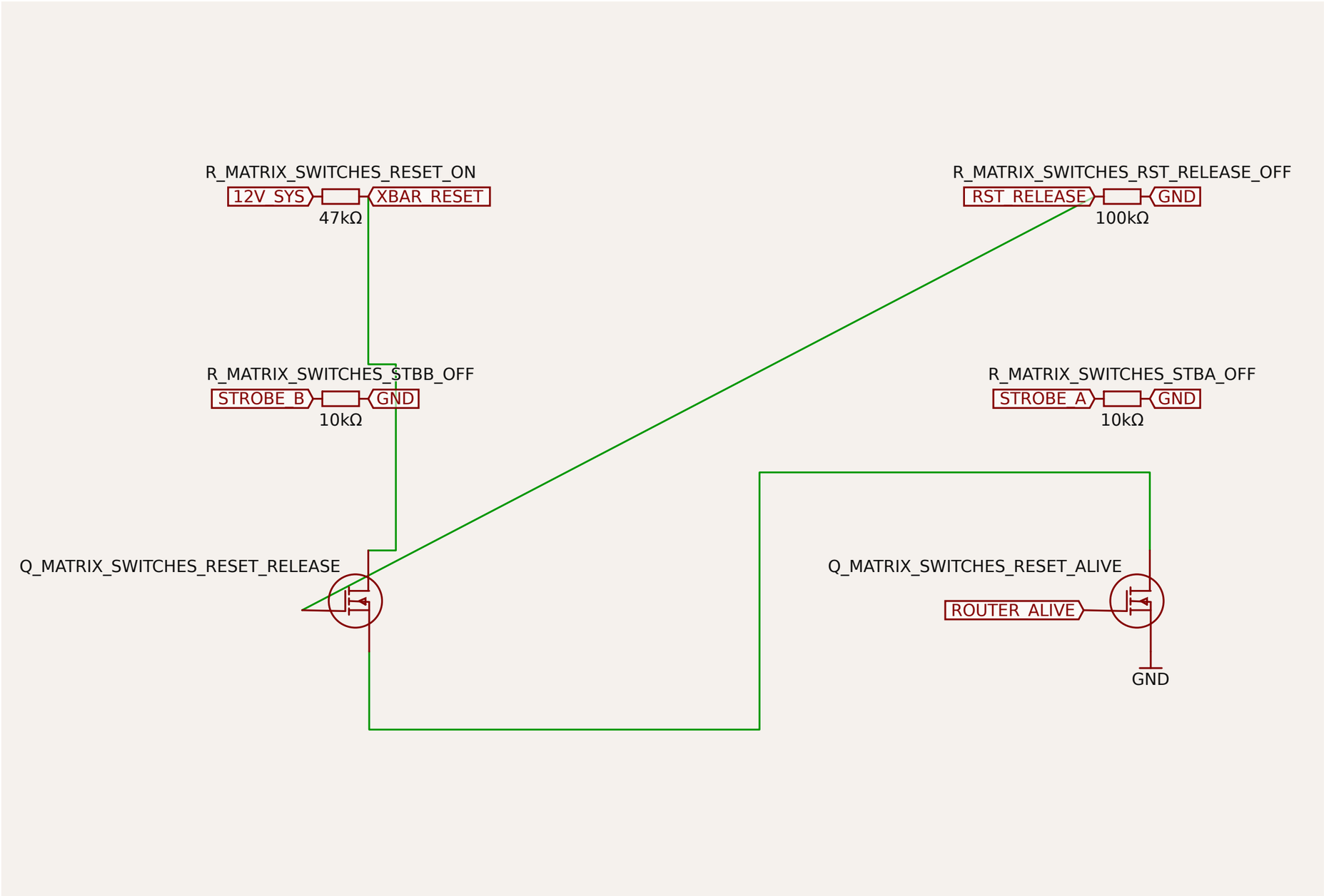
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — integration: Circuit



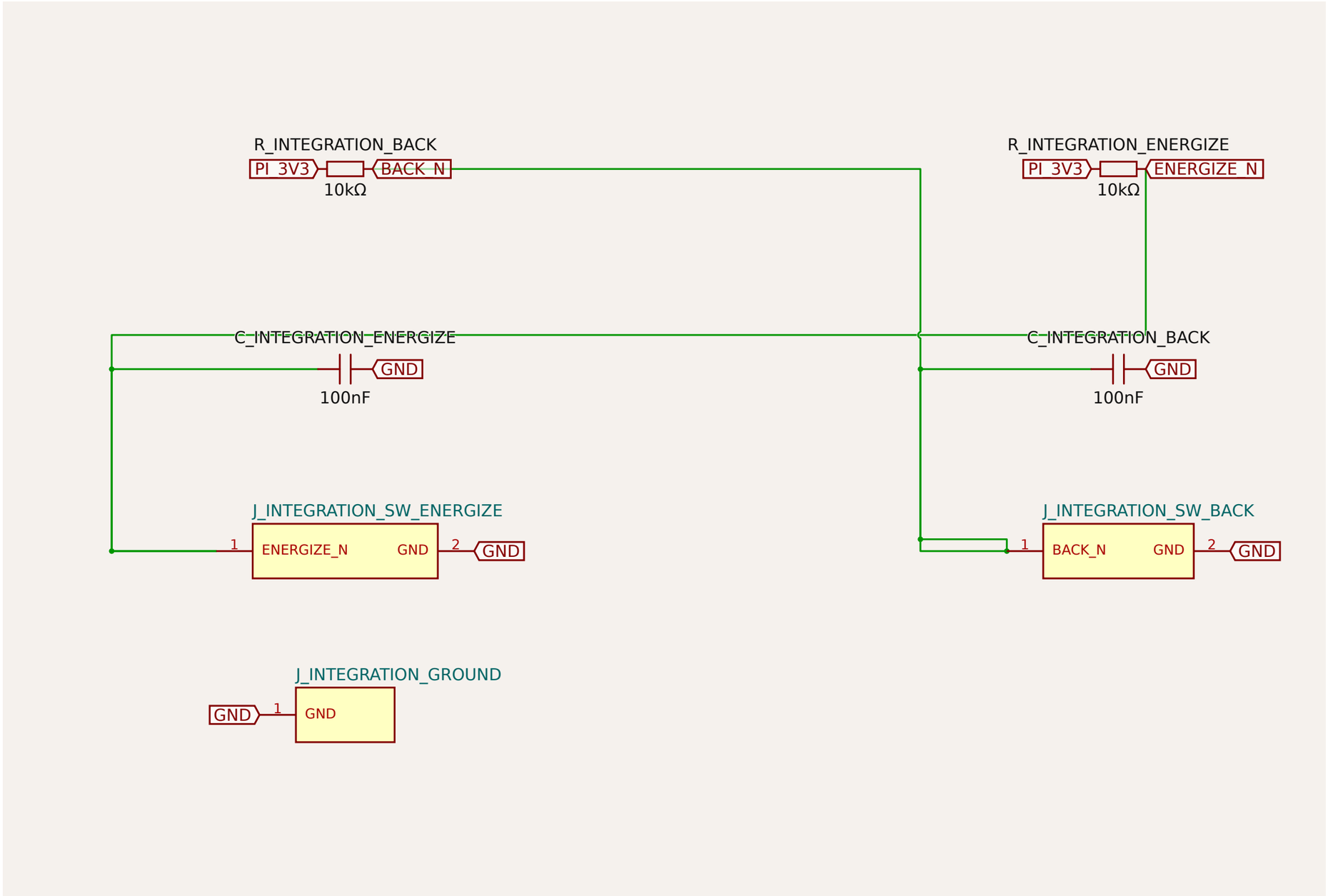
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — matrix-switches: Control



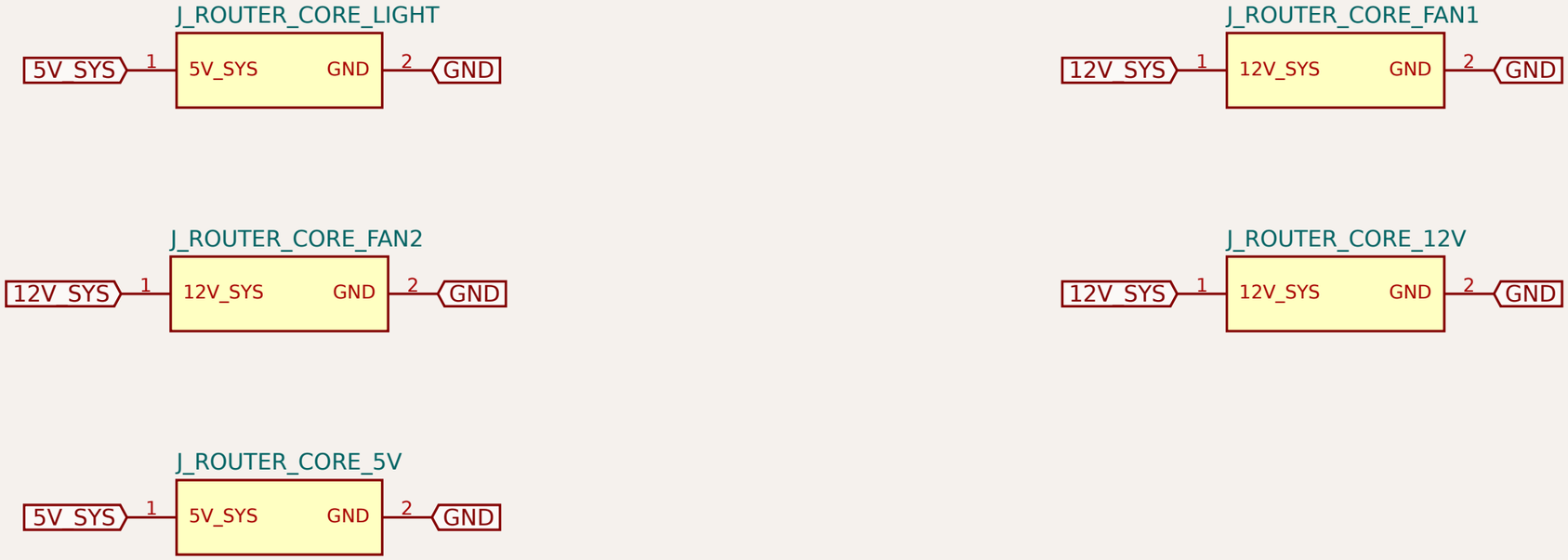
Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — integration: Buttons



Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

router-matrix — router-core: Circuit

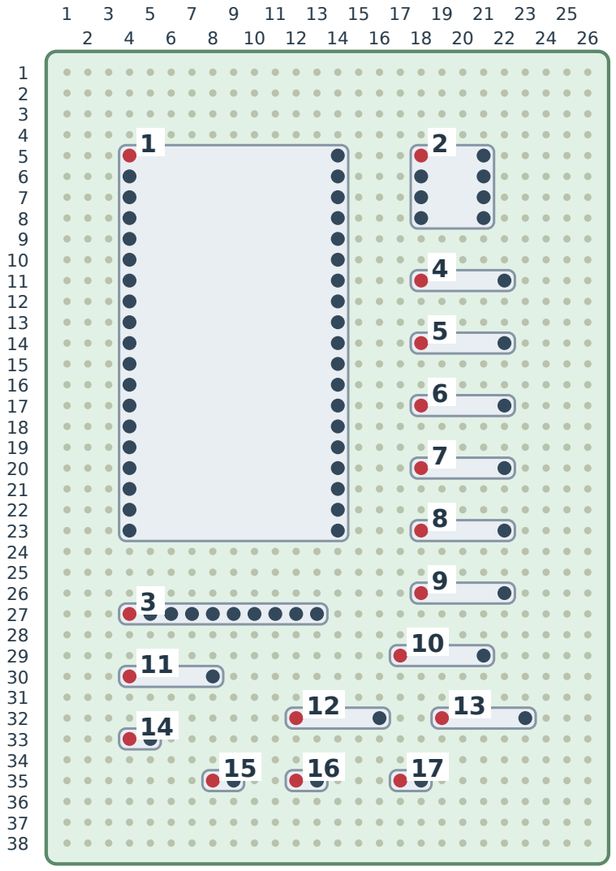


Continuous connections within a functional block; named boundaries preserve all external connections. Full model and endpoint crosswalk accompany this PDF.

Standalone early smart reference

Separate temporary fixture; not a fifth permanent perfboard

smart-early — TOP

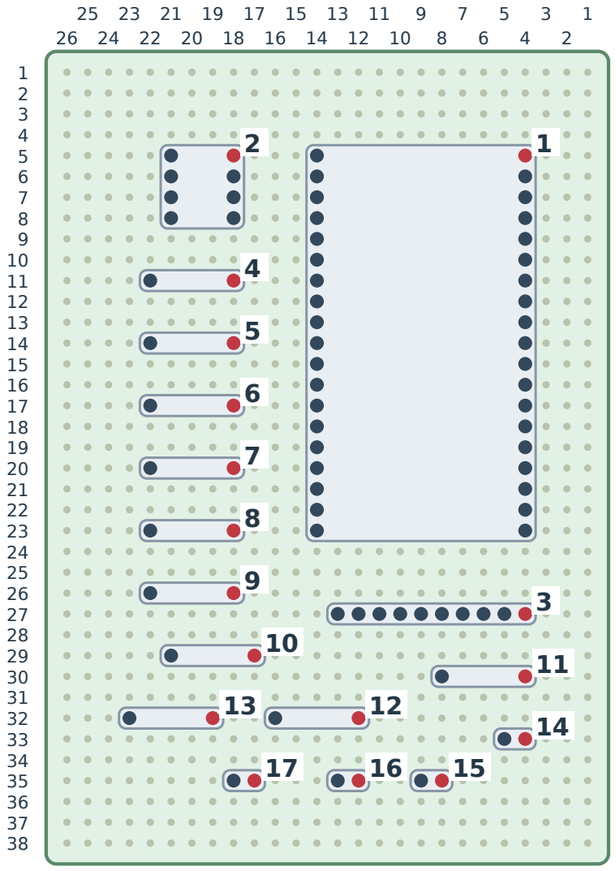


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

TOP coordinates. Crossings are not joints. See numbered ta

PROTOTYPE: physical fit and hardware tests pending.

smart-early — MIRRORED UNDERSIDE

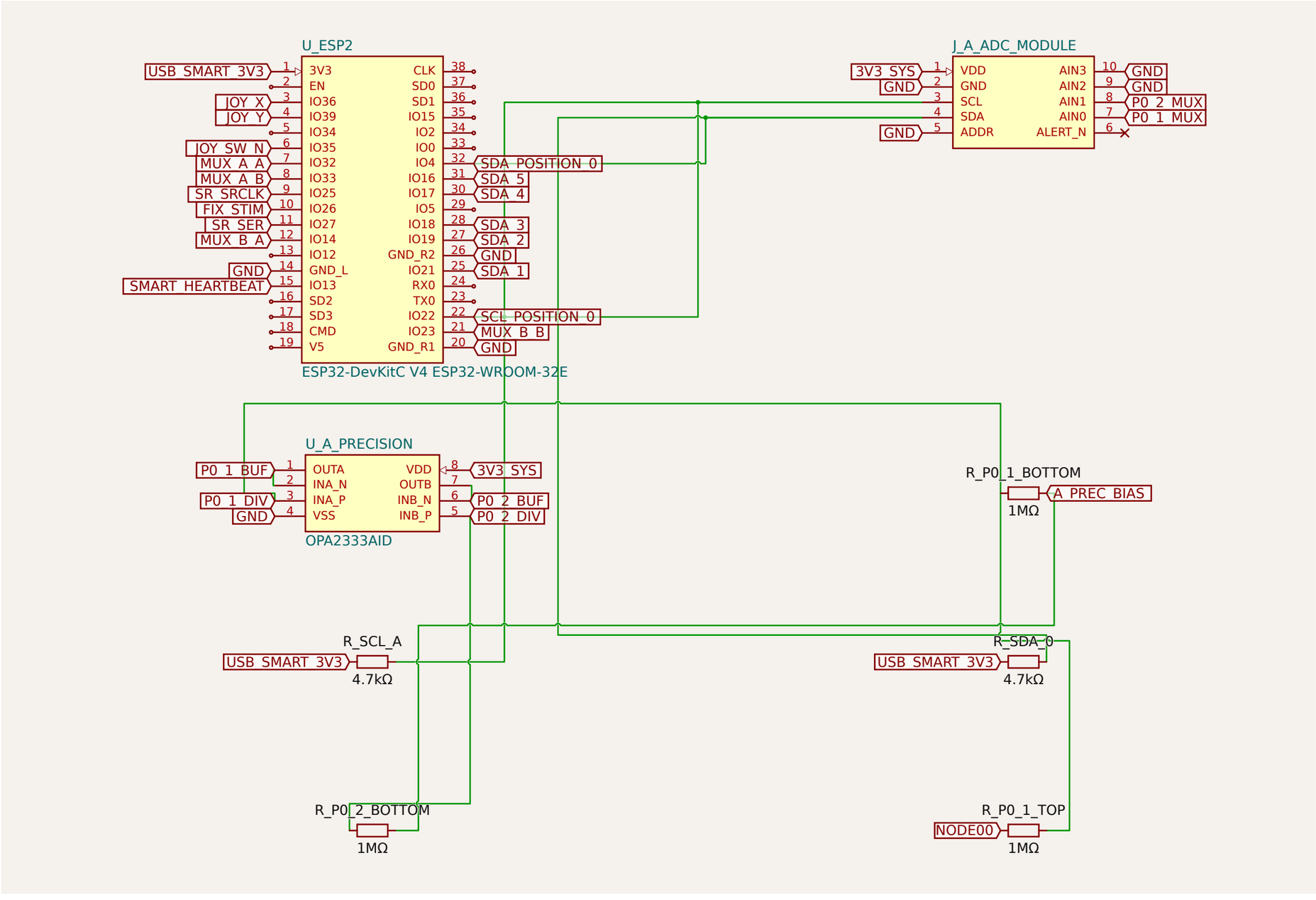


70×100 mm · 2.54 mm pitch · isolated pads · pin 1 red

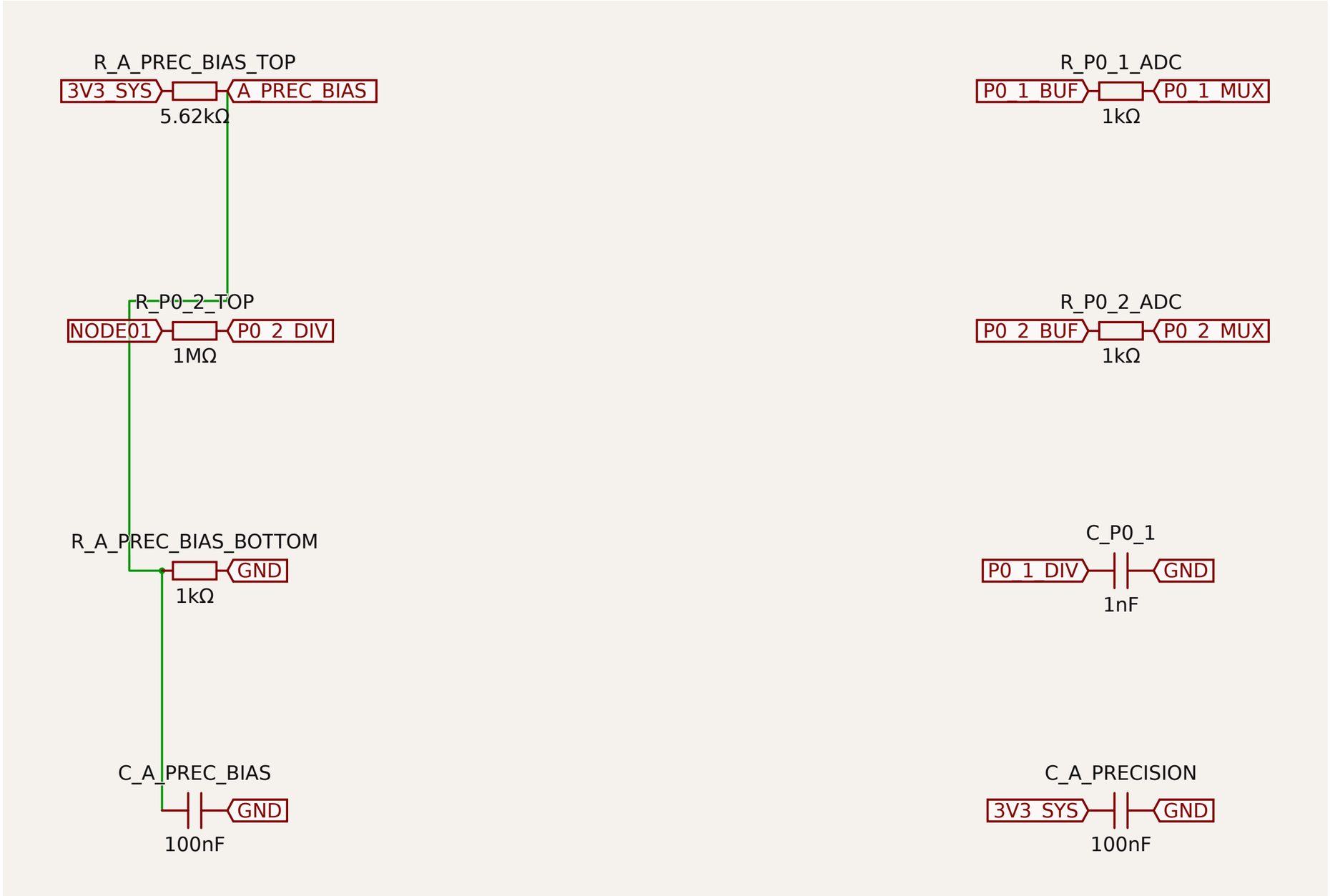
TOP coordinates. Crossings are not joints. See numbered ta

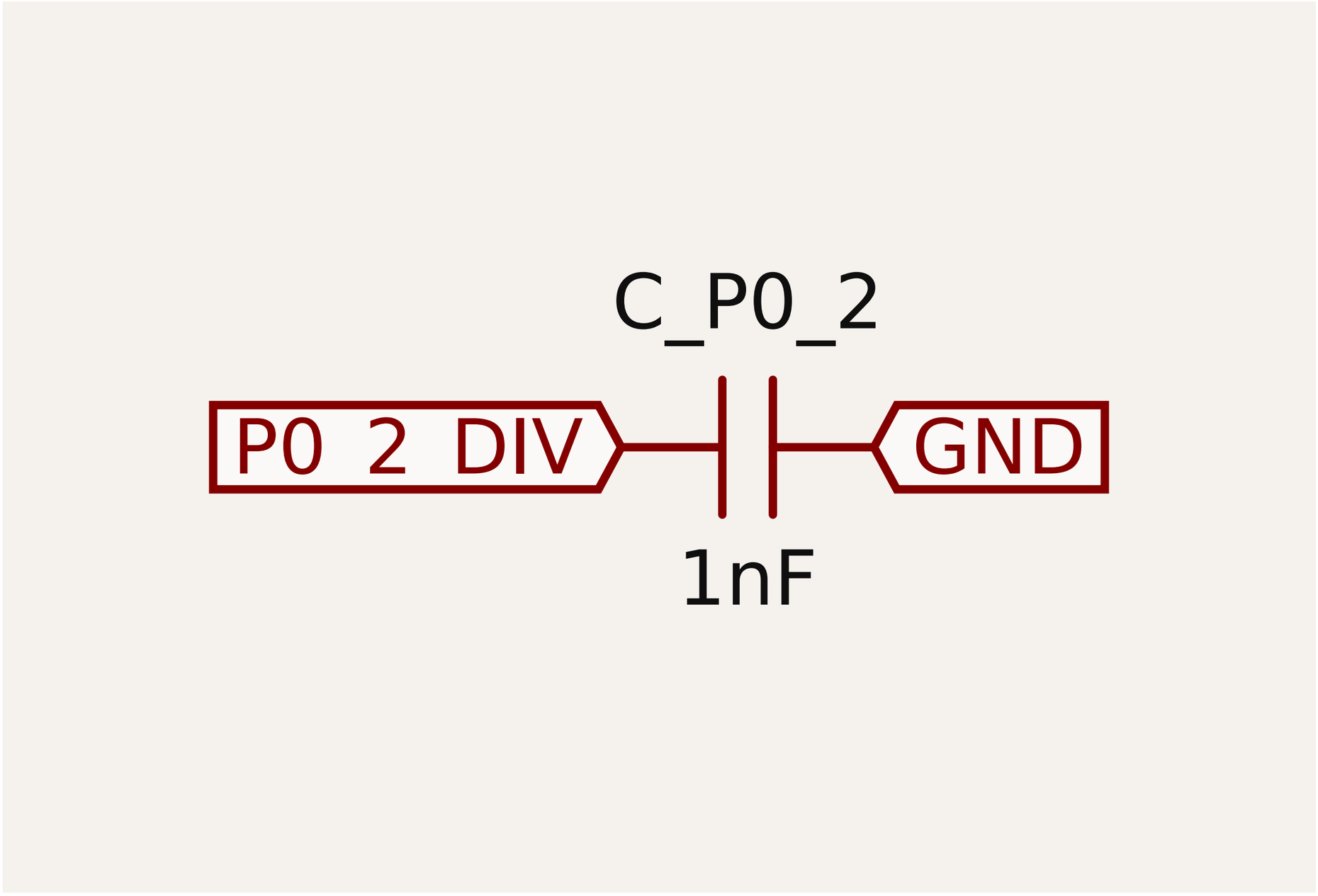
PROTOTYPE: physical fit and hardware tests pending.

Early prototype — Early reference 1

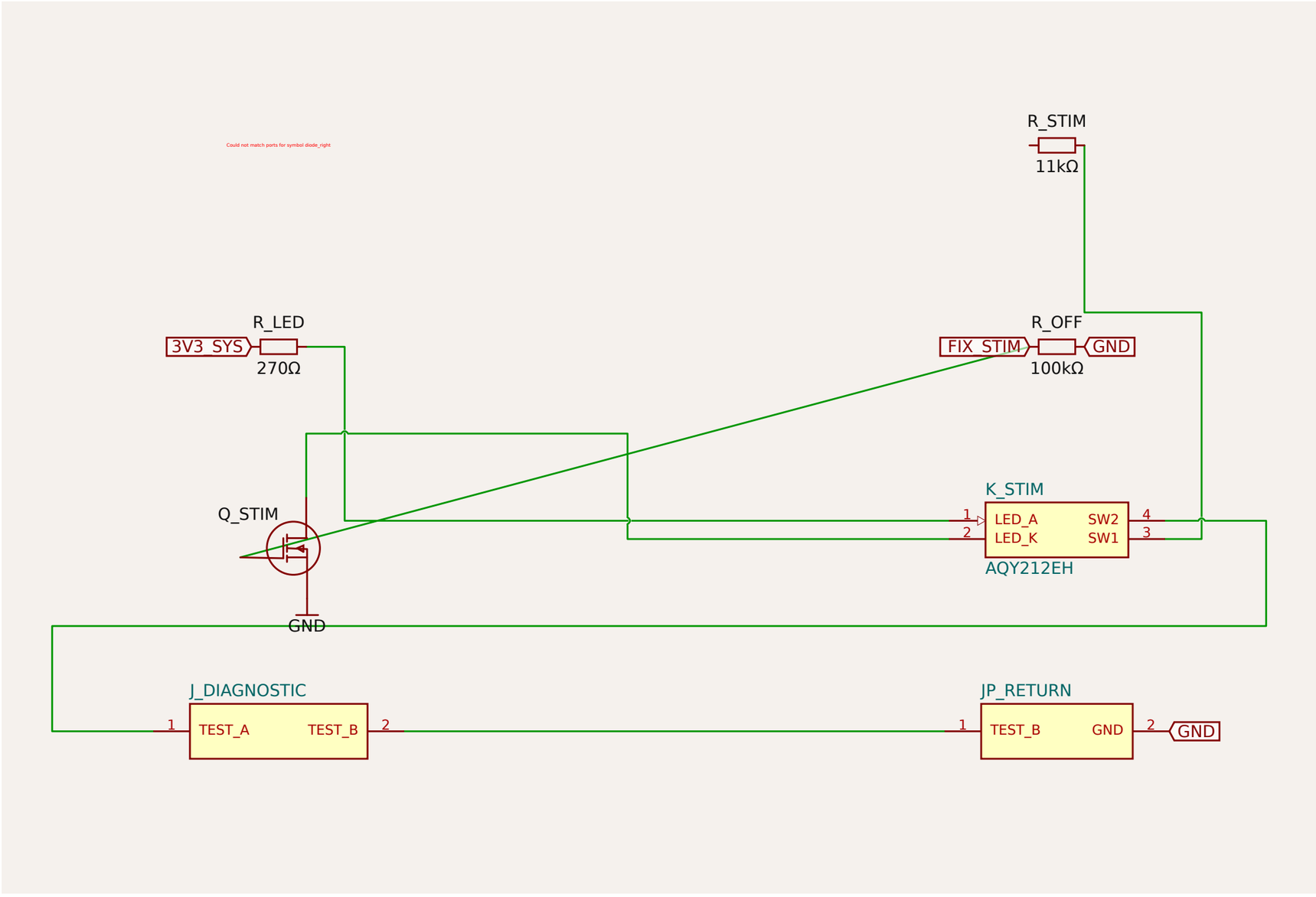


Early prototype — Early reference 2




C_P0_2
P0_2_DIV
GND
1nF

External continuity stimulus — temporary fixture



See prototype-stimulus-pins.csv for actual IC pins and net membership. FIX_STIM receives standalone GPIO26; power is verified 3.3 V; TEST_A/TEST_B and JP_RETURN are removable. Student positive must be disconnected. This drawing is a schematic, not a measured breadboard placement.